POWER ELECTRONICS

Module 3: AC Voltage Controllers, DC-AC Inverters & Utility Interface

B.Tech S5 Electrical & Electronics Engineering (EEE) · APJ Abdul Kalam Technological University

Comprehensive Master Study Guide & University Examination Preparation Manual

Revised edition — equations re-derived, worked problems re-computed numerically, figures redrawn

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AC Voltage Controllers (ACVC) / AC Regulators

Foundational Principles and Classification

An **AC Voltage Controller** (also termed an **AC Voltage Regulator**) is a line-frequency phase-controlled or integral-cycle switched converter that converts a fixed-voltage, fixed-frequency alternating current supply into a **variable-voltage, fixed-frequency AC output** without an intermediate DC stage:

$$\text{Fixed AC Input } (V_s, f) \xrightarrow{\text{AC Voltage Controller}} \text{Variable AC Output } (V_{or}, f) \tag{1}$$

Because conversion is accomplished entirely in a single stage using line-commutated thyristors or TRIACs, AC voltage controllers feature extremely high operational efficiency ($\eta > 98\%$), rugged mechanical simplicity, low initial cost, and the complete absence of bulky reactive filter banks.

Classification of AC Voltage Control Methods

AC voltage controllers are categorized into three distinct operational methodologies according to the switching mechanism employed (frequently evaluated in university examinations):

1. **Phase Angle Control (PAC):**
   - **Mechanism:** The power semiconductor switches (a pair of anti-parallel SCRs or a TRIAC) connect the load to the AC source for a controllable fraction of each supply half-cycle. Conduction is initiated at a controllable delay angle α (measured relative to the zero-crossing of the input supply voltage) and ceases naturally at current zero (β) via natural (line) commutation.
   - **Advantages:** Provides continuous, smooth, instantaneous, and infinitely variable control of RMS load voltage from 0 to V_s.
   - **Disadvantages:** The abrupt chopping of sinusoidal voltage waveforms introduces substantial low-order current harmonics (3rd, 5th, 7th) and causes the supply displacement factor to lag significantly as α increases.
   - **Primary Applications:** Universal motor speed control, light dimmers, induction motor soft-starters, industrial furnace heating.
2. **Integral Cycle Control (On-Off / Burst Firing Control):**
   - **Mechanism:** The thyristor switches connect the load to the AC mains for an integral number of complete supply cycles (n cycles), and then remain un-triggered for another integral number of cycles (m cycles). Switching is always synchronized to occur at the zero-crossing of the supply voltage ($v_s(t) = 0$).
   - **Mathematical Duty Ratio and RMS Voltage:**

$$k = \frac{n}{n+m} = \frac{T_{on}}{T_o} \implies V_{or} = V_s\sqrt{k} = V_s\sqrt{\frac{n}{n+m}} \tag{2}$$

     where $T_o = (n+m)T$ is the overall repeating period.
   - **Advantages:** Because switching transitions occur exclusively at natural voltage zero-crossings ($dv/dt \to 0$ and $di/dt \to 0$), radio frequency interference (RFI), electromagnetic interference (EMI), and switching losses are virtually zero.
   - **Disadvantages:** Sub-harmonic frequency components (frequencies below the mains supply frequency f) and line flicker are introduced, making this method completely unsuitable for motor speed control or lighting systems.
   - **Primary Applications:** Loads with very large mechanical or thermal time constants, such as industrial electric ovens, boiler heaters, and plastic extrusion mold heaters.

3. **Sequence Control of AC Voltage Controllers:**

   - **Mechanism:** Two or more parallel transformer-tapped stages of AC voltage controllers are fired in sequence. By controlling only one stage at a time while leaving others fully ON or fully OFF, the output voltage is varied with substantially reduced harmonic distortion and a significantly improved supply power factor across the entire operating range.

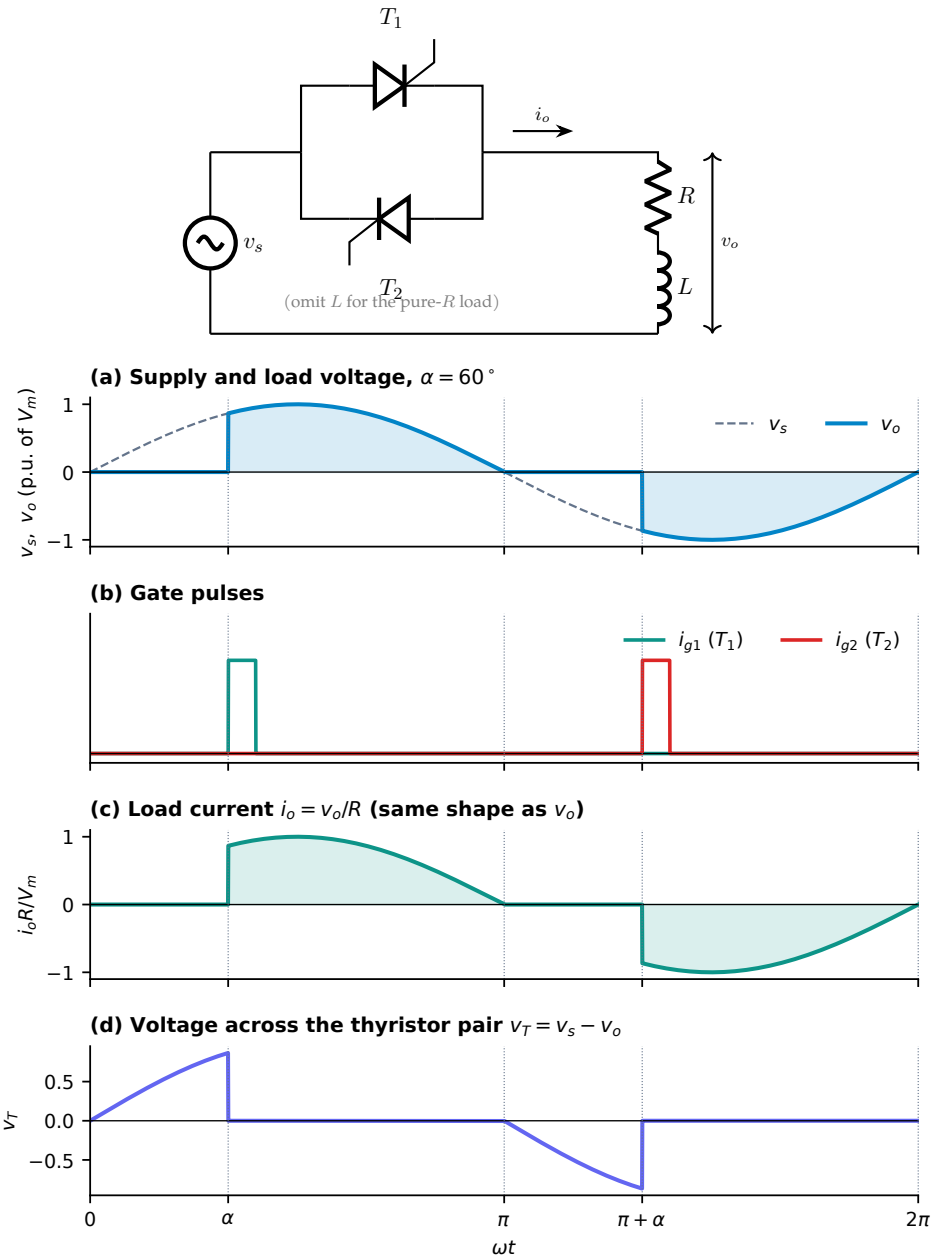


Figure 1: Single-phase full-wave AC voltage controller with resistive load, $\alpha = 60°$: (a) power circuit with two anti-parallel thyristors T_1, T_2 (add the series inductance L for the RL case); (b) waveforms of supply and load voltage, gate pulses, load current $i_o = v_o/R$ and the voltage v_T across the thyristor pair. (Redrawn; compare Hart Fig. 5-1.)

Single-Phase Full-Wave AC Voltage Controller with Pure Resistive (R) Load

Consider the circuit of Figure 1a supplied by a single-phase sinusoidal AC source $v_s(t) = V_m \sin(\omega t) = \sqrt{2}V_s \sin(\omega t)$. Two SCRs ($T_1$ and T_2) are connected in anti-parallel (back-to-back) between the AC mains and a resistive load R.

Circuit Operation Across One Full Fundamental Cycle

- **Positive Half-Cycle ($0 \le \omega t \le \pi$):** Supply voltage $v_s > 0$. Thyristor T_1 is forward-biased, while T_2 is reverse-biased ($v_{AK2} < 0$).
  - From $\omega t = 0$ to $\omega t = \alpha$, T_1 remains OFF. No current flows ($i_o = 0$), load voltage $v_o = 0$, and the full line voltage appears across T_1 ($v_{T1} = v_s$).
  - At $\omega t = \alpha$, a firing pulse is applied to the gate of T_1. T_1 turns ON instantaneously.
  - From $\omega t = \alpha$ to π, T_1 conducts load current:

$$v_o(t) = v_s(t) = V_m \sin(\omega t), \quad i_o(t) = \frac{v_s(t)}{R} = \frac{V_m}{R}\sin(\omega t) \tag{3}$$

  - At $\omega t = \pi$, the supply voltage reverses polarity. The current through T_1 drops naturally to zero ($i_o(\pi) = 0$), turning T_1 OFF via natural (line) commutation.
- **Negative Half-Cycle ($\pi \le \omega t \le 2\pi$):** Supply voltage $v_s < 0$. Thyristor T_2 is forward-biased, while T_1 is reverse-biased.
  - At $\omega t = \pi + \alpha$, a firing pulse is applied to the gate of T_2. T_2 turns ON.
  - From $\omega t = \pi + \alpha$ to 2π, T_2 conducts negative load current:

$$v_o(t) = v_s(t) = V_m \sin(\omega t), \quad i_o(t) = \frac{V_m}{R}\sin(\omega t) \quad (< 0) \tag{4}$$

  - At $\omega t = 2\pi$, $i_o(2\pi) = 0$, and T_2 turns OFF naturally.

Rigorous Mathematical Derivation of RMS Output Voltage

Because the output voltage waveform exhibits half-wave symmetry ($v_o(\omega t + \pi) = -v_o(\omega t)$), the root-mean-square (RMS) value can be computed over a half-cycle (0 to π):

$$\begin{aligned} V_{or} &= \sqrt{\frac{1}{\pi}\int_{\alpha}^{\pi} [V_m \sin(\omega t)]^2 \, d(\omega t)} = \sqrt{\frac{V_m^2}{\pi}\int_{\alpha}^{\pi} \frac{1-\cos(2\omega t)}{2} d(\omega t)} \\ &= \sqrt{\frac{V_m^2}{2\pi}\left[\omega t - \frac{\sin(2\omega t)}{2}\right]_{\alpha}^{\pi}} = \sqrt{\frac{V_m^2}{2\pi}\left[(\pi-\alpha) - \left(\frac{\sin(2\pi)-\sin(2\alpha)}{2}\right)\right]} \\ &= \sqrt{\frac{V_m^2}{2\pi}\left[(\pi-\alpha) + \frac{\sin(2\alpha)}{2}\right]} = \frac{V_m}{\sqrt{2}}\sqrt{\frac{1}{\pi}\left[(\pi-\alpha) + \frac{\sin(2\alpha)}{2}\right]} \end{aligned} \tag{5}$$

Recognizing that $V_s = V_m/\sqrt{2}$ is the RMS value of the input source:

Master Equation: RMS Output Voltage (R Load)

$$V_{or} = V_s\sqrt{1 - \frac{\alpha}{\pi} + \frac{\sin(2\alpha)}{2\pi}} \tag{6}$$

Performance Characteristics with R Load

- **RMS Load Current:** $I_{or} = \frac{V_{or}}{R} = \frac{V_s}{R}\sqrt{1 - \frac{\alpha}{\pi} + \frac{\sin(2\alpha)}{2\pi}}$.
- **Average and RMS Thyristor Currents:** Each thyristor conducts for only one half of the symmetrical period:

$$I_{T(avg)} = \frac{1}{2\pi}\int_{\alpha}^{\pi} \frac{V_m}{R}\sin(\omega t)d(\omega t) = \frac{V_m}{2\pi R}(1+\cos\alpha) = \frac{\sqrt{2}V_s}{2\pi R}(1+\cos\alpha) \tag{7}$$

$$I_{T(rms)} = \frac{I_{or}}{\sqrt{2}} = \frac{V_s}{\sqrt{2}R}\sqrt{1 - \frac{\alpha}{\pi} + \frac{\sin(2\alpha)}{2\pi}} \tag{8}$$

- **Input Power Factor (PF):** Since input current $i_s(t) = i_o(t)$, the apparent power supplied by the mains is $S = V_s I_{or}$. The active power dissipated in the load is $P_o = I_{or}^2 R = \frac{V_{or}^2}{R}$.

$$\text{PF} = \frac{P_o}{S} = \frac{V_{or}^2/R}{V_s I_{or}} = \frac{V_{or} I_{or}}{V_s I_{or}} = \frac{V_{or}}{V_s} = \sqrt{1 - \frac{\alpha}{\pi} + \frac{\sin(2\alpha)}{2\pi}} \tag{9}$$

- **Control Range:** As α varies from 0 to π ($180°$):
  - At $\alpha = 0$: $V_{or} = V_s$, PF $= 1.0$ (Full sinusoidal conduction).
  - At $\alpha = \pi/2$ ($90°$): $V_{or} = V_s/\sqrt{2} \approx 0.707 V_s$, PF ≈ 0.707.
  - At $\alpha = \pi$ ($180°$): $V_{or} = 0$, PF $= 0$ (Output completely blocked).

Single-Phase Full-Wave AC Voltage Controller with RL Load

When the load consists of an inductive impedance $Z = \sqrt{R^2 + (\omega L)^2}$ with characteristic load impedance angle $\phi = \arctan\left(\frac{\omega L}{R}\right)$, the inductance opposes sudden changes in load current. Consequently, **the current continues to flow after the source voltage reverses polarity at** $\omega t = \pi$.

Derivation of the Load Current Differential Equation

During the conduction of thyristor T_1 ($\alpha \le \omega t \le \beta$), the governing Kirchhoff's Voltage Law (KVL) differential equation is:

$$L\frac{di_o(t)}{dt} + Ri_o(t) = v_s(t) = V_m \sin(\omega t) \tag{10}$$

The total solution consists of a steady-state particular solution $i_{ss}(t)$ and a transient complementary solution $i_{tr}(t)$:

$$i_o(\omega t) = \frac{V_m}{Z}\sin(\omega t - \phi) + Ae^{-\frac{R}{\omega L}\omega t} \tag{11}$$

where $Z = \sqrt{R^2 + (\omega L)^2}$ and $\phi = \arctan(\omega L/R)$.

Applying the initial boundary condition at the firing instant: at $\omega t = \alpha$, $i_o(\alpha) = 0$:

$$0 = \frac{V_m}{Z}\sin(\alpha - \phi) + Ae^{-\frac{R}{\omega L}\alpha} \implies A = -\frac{V_m}{Z}\sin(\alpha - \phi)e^{\frac{R}{\omega L}\alpha} \tag{12}$$

Substituting A back into the current expression yields the closed-form equation for the instantaneous load current:

Instantaneous Load Current in ACVC with RL Load

$$i_o(\omega t) = \frac{V_m}{Z}\left[\sin(\omega t - \phi) - \sin(\alpha - \phi)e^{-\frac{R}{\omega L}(\omega t - \alpha)}\right], \quad \text{for } \alpha \le \omega t \le \beta \tag{13}$$

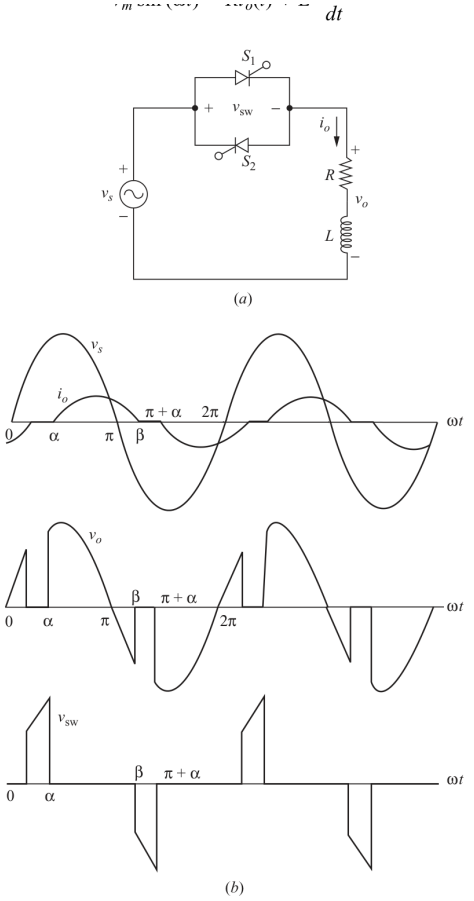


Figure 5-4 (*a*) Single-phase ac voltage controller with an *RL* load; (*b*) Typical waveforms.

Figure 2: Single-phase AC voltage controller with RL load: (a) circuit; (b) typical waveforms of the supply voltage v_s, load current i_o, load voltage v_o and switch voltage v_{sw}, showing the delay angle α, the extinction angle β and the instant $\pi + \alpha$ at which T_2 is fired. (Source: Daniel W. Hart, Fig. 5-4)

Extinction Angle (β) and Conduction Angle (γ)

At $\omega t = \beta$, the magnetic energy stored in inductor L is completely dissipated into R, and the current extinguishes to zero ($i_o(\beta) = 0$). Setting equation (13) to zero at $\omega t = \beta$ gives the transcendental equation:

$$\sin(\beta - \phi) = \sin(\alpha - \phi)e^{-\frac{R}{\omega L}(\beta - \alpha)} \tag{14}$$

The conduction angle of each thyristor is defined as $\gamma = \beta - \alpha$. Equation (14) can be expressed in terms of γ as:

$$\sin(\alpha + \gamma - \phi) = \sin(\alpha - \phi)e^{-\frac{R}{\omega L}\gamma} \tag{15}$$

Three Critical Conduction Regimes Based on Firing Angle α vs ϕ

The behavior of the RL AC voltage controller depends decisively on the relative relationship between the firing delay angle α and the load impedance phase angle ϕ:

1. **Case 1: $\alpha > \phi$ (Discontinuous Conduction Mode — Standard Operation):**
   - The current extinguishes before the next half-cycle thyristor is triggered: $\beta < \pi + \alpha$, which means the conduction angle $\gamma = (\beta - \alpha) < \pi$.
   - There is a finite zero-current interval between the extinction of T_1 and the firing of T_2.
   - The output voltage is fully controllable by varying α between ϕ and π.
2. **Case 2: $\alpha = \phi$ (Boundary / Critical Conduction Mode):**
   - Substituting $\alpha = \phi$ into equation (14):

$$\sin(\beta - \phi) = \sin(0) \cdot e^{-\frac{R}{\omega L}(\beta - \phi)} = 0 \implies \beta - \phi = \pi \implies \beta = \pi + \phi = \pi + \alpha \tag{16}$$

   - The conduction angle is exactly $\gamma = \beta - \alpha = \pi$ ($180°$).
   - Thyristor T_1 conducts for exactly $180°$. The moment T_1 extinguishes at $\omega t = \pi + \alpha$, thyristor T_2 is triggered!
   - The load current becomes a continuous, perfectly sinusoidal waveform: $i_o(t) = \frac{V_m}{Z}\sin(\omega t - \phi)$. The converter behaves as if the load is directly connected across the AC supply!
3. **Case 3: $\alpha < \phi$ (Loss of Control):**
   - The solution above would give $\beta > \pi + \alpha$, i.e. T_1 would still be carrying current when T_2 is gated at $\pi + \alpha$. T_2 is then still reverse-biased (the voltage across it is ≈ 0 while T_1 conducts).
   - **Long gate pulse:** T_2 simply takes over when T_1 current reaches zero, i.e. at $\omega t = \pi + \phi$. The firing is effectively delayed to ϕ, the load current is the continuous sinusoid $i_o = \frac{V_m}{Z}\sin(\omega t - \phi)$ and $V_{or} = V_s$ for *every* $\alpha < \phi$. Firing earlier gives no extra output.
   - **Short (narrow) gate pulse:** the pulse to T_2 is gone before T_1 stops conducting, so T_2 never fires. Only T_1 conducts, the current becomes unidirectional with a DC component, and the supply or motor transformer can saturate.
   - **Critical Exam Rule:** α must satisfy $\alpha \geq \phi$. The usable control range is $\phi \leq \alpha \leq \pi$ (with V_{or} falling from V_s at $\alpha = \phi$ to 0 at $\alpha = \pi$).

Derivation of RMS Output Voltage with RL Load

During the conduction interval $\alpha \leq \omega t \leq \beta$, the load voltage equals the supply voltage: $v_o(t) = V_m \sin(\omega t)$. Outside this interval, $v_o(t) = 0$.

$$V_{or} = \sqrt{\frac{1}{\pi}\int_{\alpha}^{\beta} [V_m \sin(\omega t)]^2 \, d(\omega t)} = \sqrt{\frac{V_m^2}{2\pi}\int_{\alpha}^{\beta} [1 - \cos(2\omega t)] \, d(\omega t)}$$
$$= \sqrt{\frac{V_m^2}{2\pi}\left[(\beta - \alpha) - \frac{\sin(2\beta) - \sin(2\alpha)}{2}\right]} \tag{17}$$

In terms of $V_s = V_m/\sqrt{2}$ and conduction angle $\gamma = \beta - \alpha$:

Master Equation: RMS Output Voltage (RL Load)

$$V_{or} = V_s\sqrt{\frac{1}{\pi}\left[(\beta - \alpha) - \frac{\sin(2\beta) - \sin(2\alpha)}{2}\right]} = V_s\sqrt{\frac{1}{\pi}\left[\gamma - \frac{\sin(2\beta) - \sin(2\alpha)}{2}\right]} \tag{18}$$

Single-Phase AC Voltage Controller with Pure Inductive (L) Load

For a purely inductive load ($R = 0 \implies \phi = 90° = \pi/2$):

$$L\frac{di_o}{dt} = V_m \sin(\omega t) \implies i_o(\omega t) = \frac{V_m}{\omega L}\int_{\alpha}^{\omega t} \sin(\omega t) d(\omega t) = \frac{V_m}{\omega L}(\cos\alpha - \cos\omega t) \tag{19}$$

The current extinguishes when $i_o(\beta) = 0$:

$$\cos\alpha - \cos\beta = 0 \implies \beta = 2\pi - \alpha \tag{20}$$

The conduction angle is:

$$\gamma = \beta - \alpha = 2\pi - 2\alpha = 2(\pi - \alpha) \tag{21}$$

- Since $\phi = \pi/2$ (90°), the control range for α is restricted to $\pi/2 \leq \alpha \leq \pi$.
- At $\alpha = \pi/2$: $\beta = 2\pi - \pi/2 = 3\pi/2$, $\gamma = \pi$ (180°). Current is continuous.
- At $\alpha = \pi$: $\beta = \pi$, $\gamma = 0$. No current flows.
- This configuration is the fundamental building block of **Thyristor-Controlled Reactors (TCR)** utilized extensively in static VAR compensators (SVC) on power transmission grids!

Gate Firing Pulse Requirements for Inductive Loads

With an inductive load, a standard short gate trigger pulse ($10 - 20\,\mu$s) cannot ensure reliable thyristor turn-on:

Gate Firing Requirements with Inductive Loads

Because $di/dt = v_s/L$ is restricted by the load inductance, the anode current rises slowly after triggering. If a short gate pulse ends before the anode current crosses the **latching current** (I_L), the thyristor will immediately turn back OFF when the gate pulse is removed!
Furthermore, due to current overlap between consecutive half-cycles, the incoming thyristor may be triggered while the outgoing thyristor is still conducting.

Mandatory Solutions:

1. **Continuous Gate Gating:** Maintain gate current throughout the desired conduction period (α to π). (Drawback: High average gate power dissipation).
2. **High-Frequency Pulse Train Gating (Carrier Frequency Modulation):** Modulate the firing signal with a $10 - 30$ kHz pulse train generated by a 555 timer or microcontroller. This reduces gate power dissipation by $> 80\%$ while allowing small, compact pulse isolation transformers!

Engineering Applications of AC Voltage Controllers

1. **Industrial and Domestic Heating:** Closed-loop temperature regulation in resistance furnaces, kilns, soldering stations, and water heaters.
2. **Lighting Control (Dimmers):** Theater stage lighting, architectural illumination, and incandescent lamp dimming.
3. **AC Motor Soft Starters:** During the starting of high-power three-phase squirrel-cage induction motors, direct-on-line (DOL) starting draws $6 - 8 \times I_{\text{rated}}$, creating severe line voltage dips. By ramping α from 150° down to ϕ, starting current is limited to $2 - 3 \times I_{\text{rated}}$.
4. **Speed Control of AC Fan and Pump Motors:** Controlling stator voltage on high-slip single-phase and three-phase induction motors provides stable variable-speed torque for centrifugal fans and water pumps.
5. **On-Load Transformer Tap Changers:** Thyristorized tap changers switch transformer winding taps under full load without mechanical arcing, contact wear, or line interruption.
6. **Static VAR Compensators (SVC):** Thyristor-Controlled Reactors (TCR) dynamically absorb variable inductive reactive power (Q) to stabilize grid voltages.

Single-Phase Switch-Mode DC-AC Voltage Source Inverters (VSI)

Fundamental Operating Principles and Bridge Topologies

An **Inverter** is a power electronic converter that transforms direct current (DC) electrical energy into alternating current (AC) electrical energy at a desired, precisely regulated voltage magnitude and frequency:

$$\text{DC Input } (V_{dc}) \xrightarrow{\text{Inverter}} \text{AC Output } (v_o(t), f_o) \tag{22}$$

In a **Voltage Source Inverter (VSI)**, the DC input source is modeled as an ideal voltage source with negligible internal impedance (stiff DC bus supported by a large electrolytic DC-link capacitor bank). The output voltage waveform is strictly determined by the switching states of the converter power semiconductors, while the output current waveform is governed by the connected load impedance.

Single-Phase Half-Bridge Inverter Configuration

The single-phase half-bridge topology (Figure 3) utilizes two fully controlled semiconductor switches (S_1, S_2) and a three-wire split DC supply with center-tap O (established by two identical series capacitors $C_1 = C_2$, each holding $V_{dc}/2$). Two anti-parallel freewheeling diodes (D_1, D_2) provide a continuous path for inductive load currents.

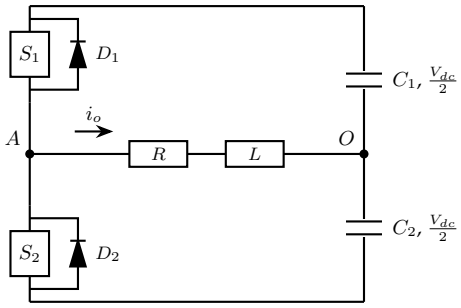


Figure 3: Single-phase half-bridge inverter: two switches with anti-parallel feedback diodes and a split DC supply formed by $C_1 = C_2$, each holding $V_{dc}/2$. The load connects the leg mid-point A to the centre tap O. (Redrawn; compare Hart Fig. 8-8.)

- **Interval 1** ($0 \leq t < T/2$)**:** Switch S_1 is turned ON. Load terminal A is connected to the positive DC rail ($+V_{dc}/2$ relative to center tap O):

$$v_o(t) = v_{AO}(t) = +\frac{V_{dc}}{2} \tag{23}$$

- **Interval 2** ($T/2 \leq t < T$)**:** Switch S_1 is turned OFF, and S_2 is turned ON. Load terminal A is connected to the negative DC rail ($-V_{dc}/2$):

$$v_o(t) = v_{AO}(t) = -\frac{V_{dc}}{2} \tag{24}$$

- **Freewheeling Diode Operation with Inductive Load:** If S_1 turns OFF while positive load current $i_o > 0$ is still flowing, inductor energy forces diode D_2 into conduction, clamping $v_o = -V_{dc}/2$ and feeding reactive energy back into the lower DC capacitor C_2 until i_o decays to zero!

- **RMS Output Voltage:**

$$V_{or} = \sqrt{\frac{1}{T}\left[\int_0^{T/2}\left(+\frac{V_{dc}}{2}\right)^2 dt + \int_{T/2}^{T}\left(-\frac{V_{dc}}{2}\right)^2 dt\right]} = \frac{V_{dc}}{2} \tag{25}$$

- **Major Limitations of the Half-Bridge Inverter:**

  1. Requires a split three-wire DC supply or large center-tapped balancing capacitors.
  2. DC bus voltage utilization is low: the maximum peak-to-peak output voltage is only V_{dc}, yielding a fundamental peak of only $\hat{V}_{o1} = \frac{2}{\pi}V_{dc} \approx 0.637V_{dc}$.
  3. Peak blocking voltage across the OFF-state switch equals the full DC supply $V_{dc} = 2 \times (V_{dc}/2)$, requiring devices rated for twice the output voltage!

Single-Phase Full-Bridge (H-Bridge) Inverter Configuration

The full-bridge inverter (Figure 4) eliminates the center-tap requirement by employing four controlled semiconductor switches (S_1, S_2, S_3, S_4) arranged in two phase legs (A and B) across a single DC source V_{dc}.

- **Diagonal Pair Conduction:**

  – When switches (S_1, S_2) are gated ON: $v_{Ao} = +V_{dc}/2$ and $v_{Bo} = -V_{dc}/2$, producing output voltage:

$$v_o(t) = v_{AB}(t) = v_{Ao} - v_{Bo} = +V_{dc} \tag{26}$$

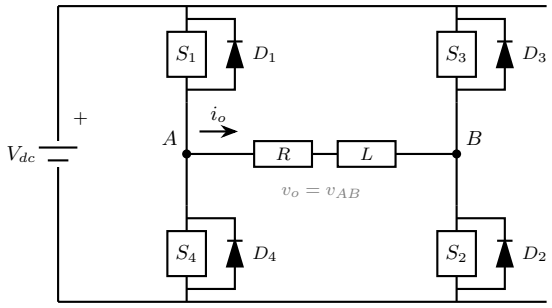


Figure 4: Single-phase full-bridge (H-bridge) inverter: leg A (S_1, S_4) and leg B (S_3, S_2), each switch with an anti-parallel diode, across one DC source V_{dc}; $v_o = v_{AB}$. (Redrawn; compare Hart Fig. 8-4.)

  - When switches (S_3, S_4) are gated ON: $v_{Ao} = -V_{dc}/2$ and $v_{Bo} = +V_{dc}/2$, producing output voltage:

$$v_o(t) = v_{AB}(t) = -V_{dc} \tag{27}$$

- **RMS Output Voltage:**

$$V_{or} = \sqrt{\frac{1}{T}\int_0^T (\pm V_{dc})^2 dt} = V_{dc} \tag{28}$$

The full-bridge produces **twice** the output voltage of the half-bridge for the same DC link voltage, yielding four times the output power into a given load resistance!

Fourier Analysis and Harmonic Spectrum of Square-Wave Inverters

The square-wave output voltage $v_o(t)$ of a full-bridge inverter alternating between $+V_{dc}$ and $-V_{dc}$ possesses **odd half-wave symmetry**:

$$v_o(\omega t) = -v_o(\omega t + \pi) \quad \text{and} \quad v_o(-\omega t) = -v_o(\omega t) \tag{29}$$

Consequently, all DC offset and all even harmonics are identically zero ($a_0 = 0$, $a_n = 0$, $b_n = 0$ for even n). The Fourier series contains only odd sine harmonics:

$$b_n = \frac{2}{\pi}\int_0^{\pi} V_{dc}\sin(n\omega t)d(\omega t) = \frac{2V_{dc}}{n\pi}\left[-\cos(n\omega t)\right]_0^{\pi} = \frac{2V_{dc}}{n\pi}\left[1-(-1)^n\right] \tag{30}$$

For odd n ($n = 1, 3, 5, 7, \ldots$), $1-(-1)^n = 2$:

Fourier Series of Single-Phase Full-Bridge Square-Wave Inverter

$$v_o(t) = \sum_{n=1,3,5,\ldots}^{\infty} \frac{4V_{dc}}{n\pi}\sin(n\omega t) = \frac{4V_{dc}}{\pi}\sin(\omega t) + \frac{4V_{dc}}{3\pi}\sin(3\omega t) + \frac{4V_{dc}}{5\pi}\sin(5\omega t) + \ldots \tag{31}$$

Key Harmonic Metrics

- **Peak Amplitude of Fundamental:** $\hat{V}_{o1} = \frac{4V_{dc}}{\pi} \approx 1.273V_{dc}$.
- **RMS Value of Fundamental:** $V_{o1(rms)} = \frac{\hat{V}_{o1}}{\sqrt{2}} = \frac{4V_{dc}}{\sqrt{2}\pi} = \frac{2\sqrt{2}}{\pi}V_{dc} \approx 0.9003V_{dc}$.
- **Harmonic Amplitudes:** The amplitude of the n-th harmonic decreases inversely with harmonic order:

$$\hat{V}_{on} = \frac{\hat{V}_{o1}}{n} \implies \frac{V_{on}}{V_{o1}} = \frac{1}{n} \tag{32}$$

Specifically: 3rd harmonic is 33.3% of fundamental, 5th harmonic is 20%, 7th harmonic is 14.3%.

- **Total Harmonic Distortion (THD) of Voltage:**

$$\text{THD}_v = \frac{\sqrt{V_{or}^2 - V_{o1(rms)}^2}}{V_{o1(rms)}} = \frac{\sqrt{V_{dc}^2 - \left(\frac{2\sqrt{2}}{\pi}V_{dc}\right)^2}}{\frac{2\sqrt{2}}{\pi}V_{dc}} = \sqrt{\frac{\pi^2}{8} - 1} \approx \sqrt{1.2337 - 1} = 0.4834 \implies 48.34\% \tag{33}$$

A total harmonic distortion of 48.34% is severe and unacceptable for sensitive AC loads or utility grid interconnection, motivating the necessity of Pulse Width Modulation (PWM)!

Square-Wave Inverter with an RL Load: Current Waveform and Conduction Sequence

With a series RL load, time constant $\tau = L/R$ and period $T = 1/f_o$, the load current is found piece-wise. For $0 \le t < T/2$ the full bridge applies $+V_{dc}$:

$$i_o(t) = \frac{V_{dc}}{R} + \left(I_0 - \frac{V_{dc}}{R}\right) e^{-t/\tau} \tag{34}$$

In steady state half-wave symmetry demands $i_o(T/2) = -I_0$, which fixes the starting value:

Steady-State Load Current of the Square-Wave Inverter (RL load)

$$I_0 = -\frac{V_{dc}}{R}\frac{1 - e^{-T/2\tau}}{1 + e^{-T/2\tau}} = -\frac{V_{dc}}{R}\tanh\left(\frac{T}{4\tau}\right), \qquad I_{\max} = -I_0$$
$$t_0 = \tau \ln\left[1 + \tanh\left(\frac{T}{4\tau}\right)\right] \tag{35}$$

t_0 is the instant at which the current crosses zero. Between 0 and t_0 the current is still negative although $v_o = +V_{dc}$: it flows through the **feedback diodes** and returns energy to the DC source. This is why *every* inverter switch needs an anti-parallel diode.

Table 1: Conduction sequence of the full-bridge square-wave inverter with an RL load (half-bridge: same sequence $D_1 \to S_1 \to D_2 \to S_2$ with $\pm V_{dc}/2$).

Interval	Gated pair	v_o	i_o	Conducting devices
$0 < \omega t < \omega t_0$	S_1, S_2	$+V_{dc}$	< 0	D_1, D_2 (energy fed back to the source)
$\omega t_0 < \omega t < \pi$	S_1, S_2	$+V_{dc}$	> 0	S_1, S_2 (source supplies the load)
$\pi < \omega t < \pi + \omega t_0$	S_3, S_4	$-V_{dc}$	> 0	D_3, D_4 (energy fed back)
$\pi + \omega t_0 < \omega t < 2\pi$	S_3, S_4	$-V_{dc}$	< 0	S_3, S_4

By superposition, each voltage harmonic of Eq. (31) drives a current harmonic through the load impedance at that frequency:

$$i_o(t) = \sum_{n=1,3,5,\ldots} \frac{4V_{dc}}{n\pi Z_n}\sin(n\omega t - \phi_n), \qquad Z_n = \sqrt{R^2 + (n\omega L)^2}, \quad \phi_n = \arctan\frac{n\omega L}{R} \tag{36}$$

Since Z_n grows with n, current harmonics fall off faster than $1/n$ and the current is far smoother than the voltage: the load inductance is a low-pass filter. Real power is delivered only by like-order voltage and current harmonics, $P = \sum_n I_{n(rms)}^2 R$.

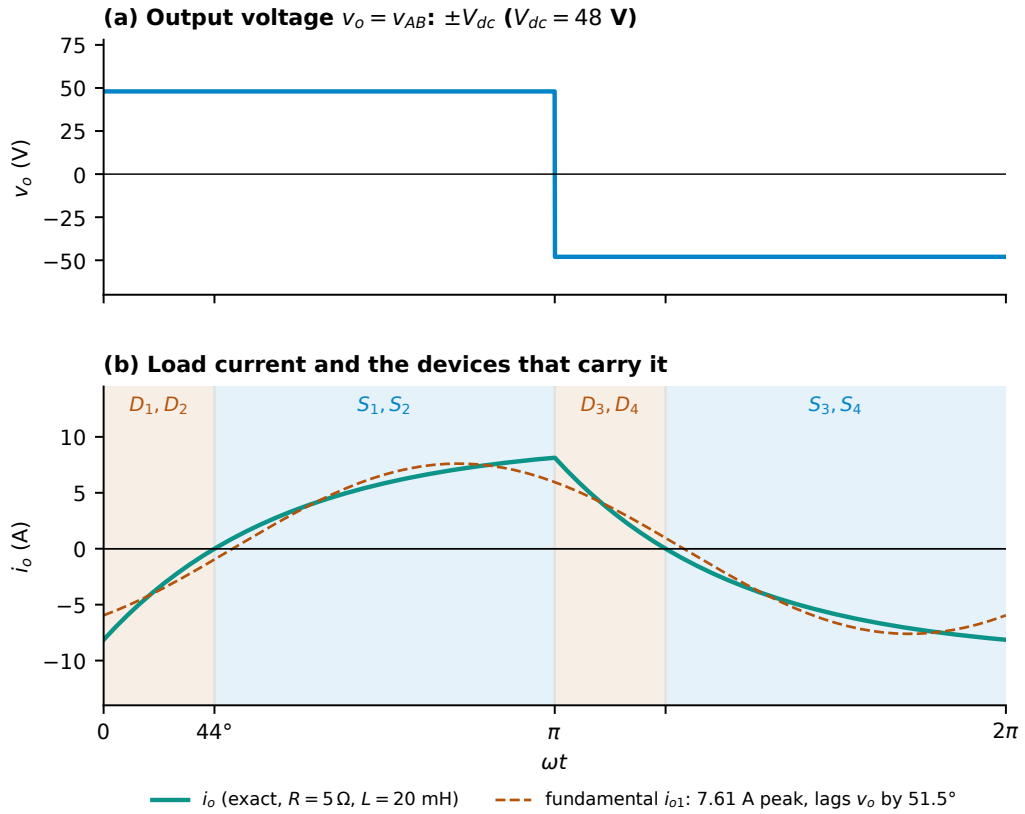


Figure 5: Full-bridge square-wave inverter, $V_{dc} = 48$ V, $R = 5\,\Omega$, $L = 20$ mH, 50 Hz (the data of Problem 3): (a) output voltage; (b) exact load current with the devices that carry it, and its fundamental component.

Single-Pulse-Width Modulation

In **single-pulse modulation** there is only **one pulse per half-cycle**, centred on the middle of the half-cycle, and the output voltage is controlled by the pulse width δ ($0 \le \delta \le \pi$). For the full bridge, $v_o = +V_{dc}$ for $\frac{\pi-\delta}{2} < \omega t < \frac{\pi+\delta}{2}$, $v_o = -V_{dc}$ in the corresponding interval of the negative half-cycle and $v_o = 0$ elsewhere (the zero state is obtained by turning on S_1, S_3 or S_4, S_2, which short-circuits the load).

Single-Pulse PWM of the Full-Bridge Inverter

$$V_{or} = \sqrt{\frac{2}{2\pi}\int_{(\pi-\delta)/2}^{(\pi+\delta)/2} V_{dc}^2\, d(\omega t)} = V_{dc}\sqrt{\frac{\delta}{\pi}} \tag{37}$$

$$v_o(t) = \sum_{n=1,3,5,\ldots} \frac{4V_{dc}}{n\pi}\sin\!\left(\frac{n\pi}{2}\right)\sin\!\left(\frac{n\delta}{2}\right)\sin(n\omega t) \tag{38}$$

The fundamental peak is $\hat{V}_{o1} = \frac{4V_{dc}}{\pi}\sin\frac{\delta}{2}$ (square wave at $\delta = \pi$). The n-th harmonic is eliminated when $\sin\frac{n\delta}{2} = 0$, i.e. $\delta = \frac{2\pi}{n}$: $\delta = 120°$ removes the 3rd harmonic and $\delta = 72°$ the 5th. **Disadvantage:** only one harmonic can be eliminated at a time, and the fundamental falls when δ is reduced (at $\delta = 120°$ it is 0.866 of the square-wave value). This motivates multiple-pulse (sinusoidal) PWM, which pushes all low-order harmonics up to the carrier frequency.

Sinusoidal Pulse Width Modulation (SPWM)

In **Sinusoidal Pulse Width Modulation (SPWM)**, the inverter switches are turned ON and OFF multiple times per half-cycle at a high switching frequency. The widths of the gating pulses are modulated proportionally to the instantaneous amplitude of a sinusoidal reference wave.

Fundamental Definitions and Modulation Parameters

1. **Modulating / Reference Wave (v_{control} or v_{sine}):** A low-frequency sinusoidal voltage whose frequency f_o defines the fundamental output frequency of the inverter, and whose peak amplitude $\hat{V}_{\text{control}}$ dictates the output voltage magnitude.
2. **Carrier Wave (v_{tri}):** A high-frequency symmetrical triangular waveform of frequency f_s (or f_c) and peak amplitude $\hat{V}_{\text{tri}}$.
3. **Amplitude Modulation Index (m_a):**

   Amplitude Modulation Index

   $$m_a = \frac{\hat{V}_{\text{control}}}{\hat{V}_{\text{tri}}} \tag{39}$$

   m_a governs the amplitude of the fundamental output voltage component.
4. **Frequency Modulation Ratio (m_f):**

   Frequency Modulation Ratio

   $$m_f = \frac{f_s}{f_o} = \frac{f_{\text{carrier}}}{f_{\text{reference}}} \tag{40}$$

   m_f determines the frequency at which switching harmonics appear in the output spectrum.

PWM with Bipolar Voltage Switching

In bipolar voltage switching, the diagonally opposite switch pairs (S_1, S_2) and (S_3, S_4) in the full bridge are switched simultaneously in complementary fashion:

- When $v_{\text{control}} > v_{\text{tri}}$: Switches S_1 and S_2 are ON $\implies v_o(t) = +V_{dc}$.
- When $v_{\text{control}} < v_{\text{tri}}$: Switches S_3 and S_4 are ON $\implies v_o(t) = -V_{dc}$.

Because the output voltage transitions directly between the two extreme polarities ($+V_{dc}$ and $-V_{dc}$), this scheme is termed **bipolar switching**.

Harmonic Characteristics of Bipolar PWM

- **Fundamental Peak Voltage in Linear Range ($m_a \leq 1.0$):**

$$\hat{V}_{o1} = m_a V_{dc} \implies V_{o1(rms)} = \frac{m_a V_{dc}}{\sqrt{2}} \approx 0.707 m_a V_{dc} \tag{41}$$

The fundamental output voltage is strictly linearly proportional to the modulation index m_a.

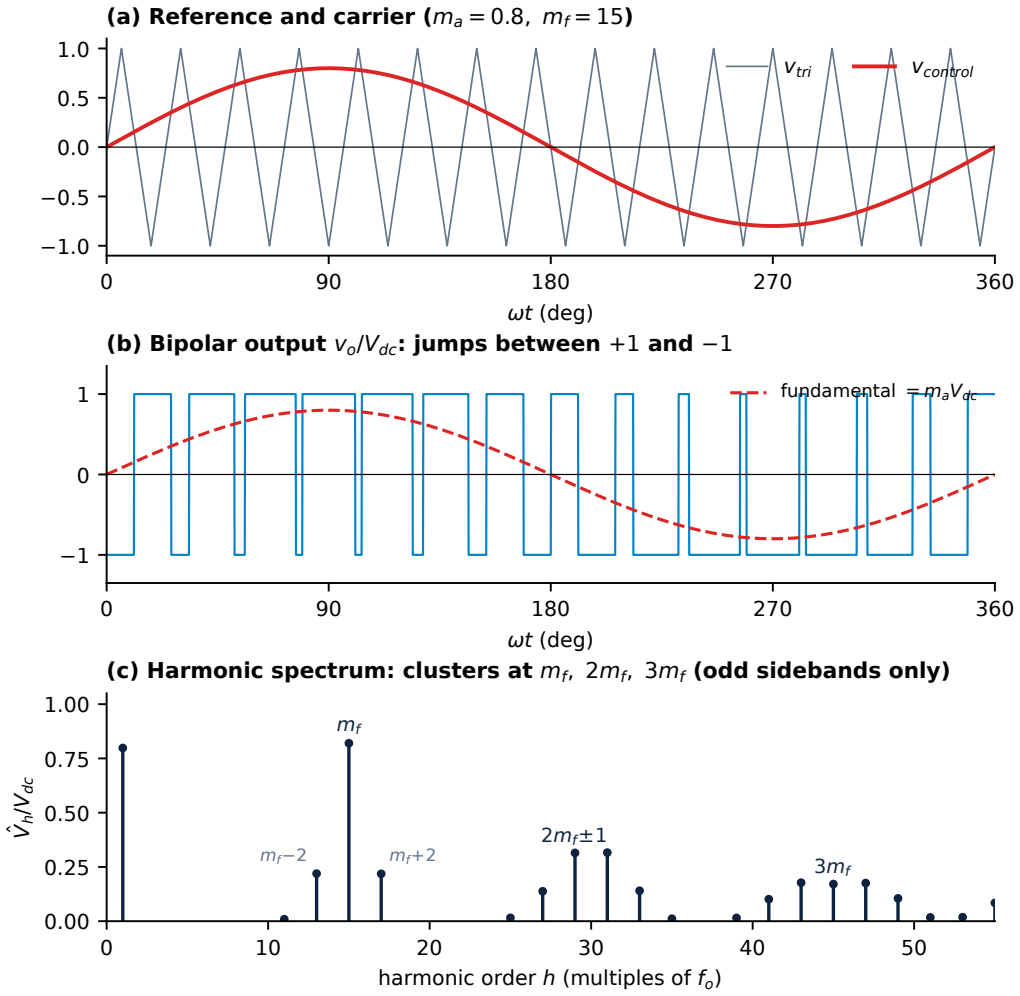


Figure 6: Bipolar PWM, $m_a = 0.8$, $m_f = 15$ (computed): (a) reference and triangular carrier; (b) output v_o/V_{dc} switching between $+1$ and -1, dashed: fundamental of amplitude $m_a V_{dc}$; (c) harmonic spectrum (FFT of the simulated waveform) with clusters around $m_f, 2m_f, 3m_f$. Compare Mohan Fig. 8-5 (one leg).

- **Harmonic Sidebands:** Harmonics appear as sidebands clustered around the carrier frequency and its integer multiples:

$$f_h = (j \cdot m_f \pm k)\, f_o \tag{42}$$

where the harmonic order is $h = j{\cdot}m_f{\pm}k$. For odd m_f, k is *even* when j is odd ($m_f,\ m_f{\pm}2,\ 3m_f{\pm}2, \dots$) and *odd* when j is even ($2m_f{\pm}1,\ 2m_f{\pm}3, \dots$), so all even harmonics cancel.

- **Selection Rules for m_f:**
  - m_f should be chosen as an **odd integer**. This establishes **odd half-wave symmetry** ($v_o(\omega t+\pi) = -v_o(\omega t)$), which completely eliminates all even harmonics from the output spectrum!
  - If m_f is an odd integer, the dominant harmonics occur at m_f and $m_f \pm 2$.
  - Synchronous PWM (m_f = integer) is essential for $m_f < 21$ to prevent sub-harmonic frequencies. For $m_f > 21$, asynchronous PWM can be utilized.

Table 2: Generalised harmonics of bipolar PWM for large odd m_f: $\hat{V}_{oh}/V_{dc}$ for the full bridge (identical to $\hat{V}_{Ao,h}/(V_{dc}/2)$ of one leg). Values reproduce Mohan Table 8-1 and were re-checked by FFT simulation; entries below 0.01 are omitted. RMS value of a harmonic = tabulated value $\times V_{dc}/\sqrt{2}$.

$h \setminus m_a$	0.2	0.4	0.6	0.8	1.0
1 (fundamental)	0.2	0.4	0.6	0.8	1.0
m_f	1.242	1.15	1.006	0.818	0.601
$m_f \pm 2$	0.016	0.061	0.131	0.220	0.318
$2m_f \pm 1$	0.190	0.326	0.370	0.314	0.181
$2m_f \pm 3$	—	0.024	0.071	0.139	0.212
$2m_f \pm 5$	—	—	—	0.013	0.033
$3m_f$	0.335	0.123	0.083	0.171	0.113
$3m_f \pm 2$	0.044	0.139	0.203	0.176	0.062
$3m_f \pm 4$	—	0.012	0.047	0.104	0.157

PWM with Unipolar Voltage Switching

In unipolar voltage switching, the two inverter legs are controlled independently by comparing the carrier wave v_{tri} with two reference sine waves that are $180°$ out of phase:

- **Leg A Control:** v_{control} is compared with v_{tri}:
  - If $v_{\text{control}} > v_{\text{tri}}$: Switch S_1 is ON ($v_{Ao} = +V_{dc}/2$).
  - If $v_{\text{control}} < v_{\text{tri}}$: Switch S_4 is ON ($v_{Ao} = -V_{dc}/2$).
- **Leg B Control:** $-v_{\text{control}}$ is compared with v_{tri}:
  - If $-v_{\text{control}} > v_{\text{tri}}$: Switch S_3 is ON ($v_{Bo} = +V_{dc}/2$).
  - If $-v_{\text{control}} < v_{\text{tri}}$: Switch S_2 is ON ($v_{Bo} = -V_{dc}/2$).

The Remarkable Advantages of Unipolar PWM

During each switching event, the output voltage transitions between **three voltage levels**: $+V_{dc}$, 0, and $-V_{dc}$:

- During the positive half-cycle of v_{control}, $v_o(t)$ switches between 0 and $+V_{dc}$.
- During the negative half-cycle of v_{control}, $v_o(t)$ switches between 0 and $-V_{dc}$.

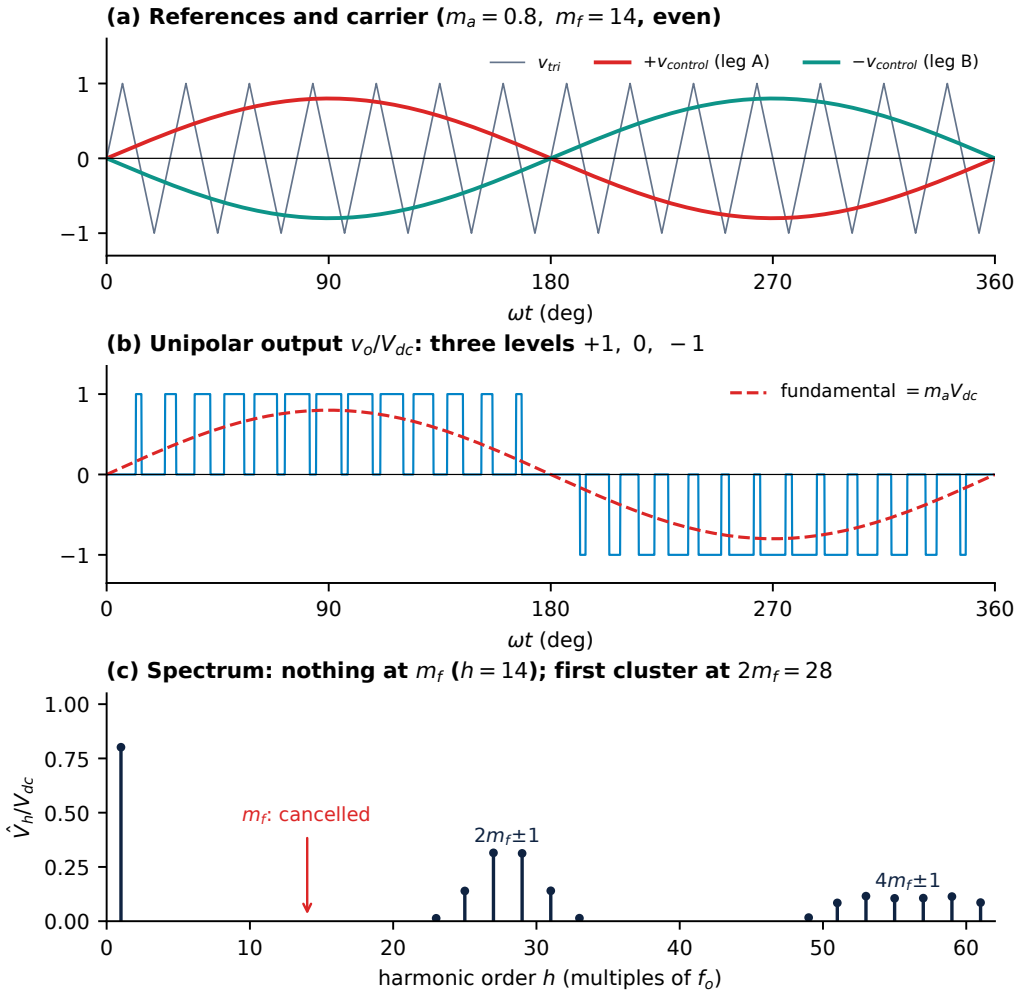


Figure 7: Unipolar PWM, $m_a = 0.8$, $m_f = 14$ (computed): (a) the two references $\pm v_{control}$ against one carrier; (b) three-level output v_o/V_{dc} (+1, 0, −1); (c) spectrum: nothing at m_f (marked), first cluster at $2m_f$. The $2m_f \pm 1$, $2m_f \pm 3$ amplitudes equal the corresponding entries of Table 2. Compare Mohan Fig. 8-15.

Why Unipolar Switching Outperforms Bipolar Switching

1. **Doubling of Effective Switching Frequency:** The output voltage pulses appear at **twice the carrier frequency** $(2f_s)$. The fundamental output voltage equation remains identical:

$$\hat{V}_{o1} = m_a V_{dc} \quad (0 \leq m_a \leq 1.0) \tag{43}$$

2. **Cancellation of Harmonics at m_f:** Harmonics centered around $m_f, 3m_f, 5m_f, \ldots$ are completely cancelled out! The lowest-order harmonic sidebands appear centered around $2m_f$:

$$h = 2m_f \pm 1, \quad 2m_f \pm 3 \tag{44}$$

3. **Dramatically Smaller Output Filter:** Because the lowest harmonic frequency is doubled $(2f_s)$, the required inductance and capacitance of the output low-pass filter can each be roughly halved (the LC product falls to about one quarter), slashing converter weight, volume, and cost!

4. **Reduced Current Ripple:** Load current ripple is substantially smaller compared to bipolar PWM for identical device switching frequencies.

Table 3: Comprehensive Comparison: Bipolar vs. Unipolar PWM Schemes

Feature	Bipolar PWM	Unipolar PWM
Output Voltage Levels	2 levels: $+V_{dc}, -V_{dc}$	3 levels: $+V_{dc}, 0, -V_{dc}$
Leg Control	Legs A and B switched simultaneously	Legs A and B switched independently
Carrier References	Single sine wave v_{control}	Complementary pair: $+v_{\text{control}}, -v_{\text{control}}$
Selection of m_f	m_f must be an odd integer	m_f is typically chosen as an even integer
Effective Output Frequency	f_s (Carrier frequency)	$2f_s$ (Double carrier frequency)
Dominant Harmonics	Clustered around $m_f, 2m_f, 3m_f$	Absent at m_f; first appear around $2m_f$
Output Filter Size	Large	Significantly smaller (L and C each $\approx$ halved, LC product $\approx$ one quarter)
Switching Stress (dv/dt)	Steps of $2V_{dc}$ across load terminals	Steps of V_{dc} across load terminals
Control Complexity	Minimal	Moderately higher

Operating Modes: Linear Modulation, Overmodulation, and Square-Wave

The relationship between the fundamental output voltage $\hat{V}_{o1}$ and the modulation index m_a spans three distinct operational regimes:

1. **Linear Modulation Region ($0 \leq m_a \leq 1.0$):**
   - The peak of the modulating sine wave is less than or equal to the triangular carrier peak ($\hat{V}_{\text{control}} \leq \hat{V}_{\text{tri}}$).
   - The fundamental peak output voltage varies strictly linearly with m_a:

$$\hat{V}_{o1} = m_a V_{dc} \tag{45}$$

   - The harmonic frequencies are independent of m_a, and harmonic amplitudes are directly proportional to m_a.

2. **Overmodulation Region ($m_a > 1.0$):**

- The reference amplitude exceeds the carrier peak. Around the sine-wave peaks the reference stays above the carrier for several carrier cycles, so **pulses are dropped**: the switch stays ON (or OFF) without pulsing.
- $\hat{V}_{o1}$ is no longer proportional to m_a: it rises ever more slowly from V_{dc} at $m_a = 1$ towards $\frac{4V_{dc}}{\pi}$ (Fig. 8). In Mohan's one-leg notation, $\frac{V_{dc}}{2} < \hat{V}_{Ao1} < \frac{4}{\pi}\frac{V_{dc}}{2}$.
- Low-order harmonics (3rd, 5th, 7th, ...) reappear and the spectrum becomes sensitive to m_f (Mohan Fig. 8-7), so synchronous PWM is recommended. Overmodulation is avoided in UPS systems (strict distortion limits) but normally used in induction-motor drives to extract more voltage.

3. **Square-Wave Operation (m_a very large):**

- The pulses merge into one pulse per half-cycle. The limit $\frac{4V_{dc}}{\pi}$ is approached asymptotically: our simulation ($m_f = 39$) already reaches 98% of it at $m_a \approx 3.2$, which is why books quote a practical threshold of "$m_a \gtrsim 3$–4" rather than a sharp value.
- The fundamental saturates at

$$\hat{V}_{o1,\max} = \frac{4V_{dc}}{\pi} \approx 1.273 V_{dc} \tag{46}$$

  i.e. 27% more than the linear-range limit V_{dc}. Each switch conducts 180° and there is no harmonic reduction ($\text{THD}_v = 48.3\%$).
- The output is independent of m_a and can only be changed through the DC link voltage V_{dc}.

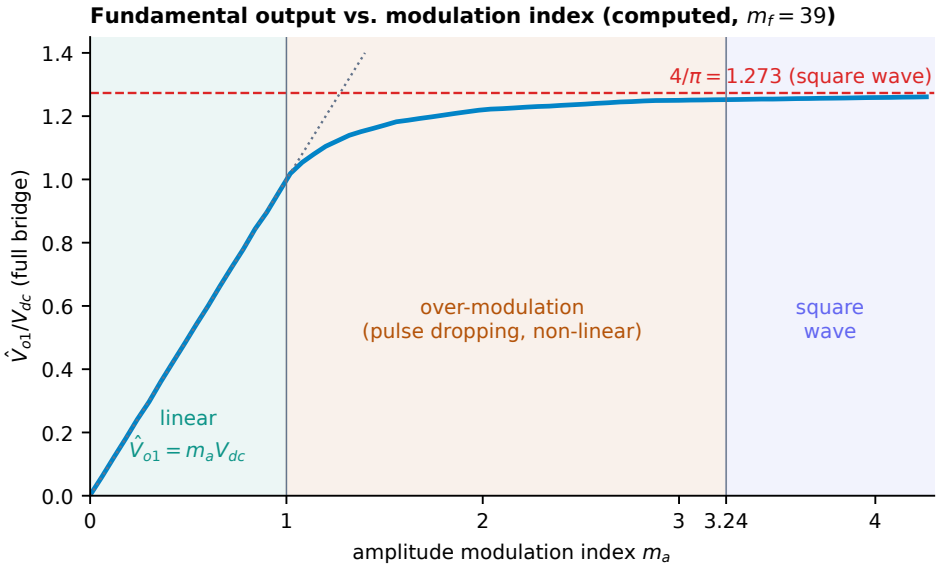


Figure 8: Fundamental amplitude of the bipolar full-bridge output versus m_a (computed, $m_f = 39$): linear region, over-modulation and approach to the square-wave limit $4/\pi$. Compare Mohan Fig. 8-8.

The Critical Need for Blanking Time (Dead-Time)

In practical bridge inverters, power semiconductor switches (IGBTs, MOSFETs) do not turn OFF instantaneously due to stored minority carrier recombination and gate charge discharging times ($t_{off} > t_{on}$).

The Shoot-Through Catastrophe

If the turn-ON command is given to the upper switch (S_1) at the same instant as the turn-OFF command to the lower switch (S_4) of the same leg, both devices conduct together while the outgoing device is still

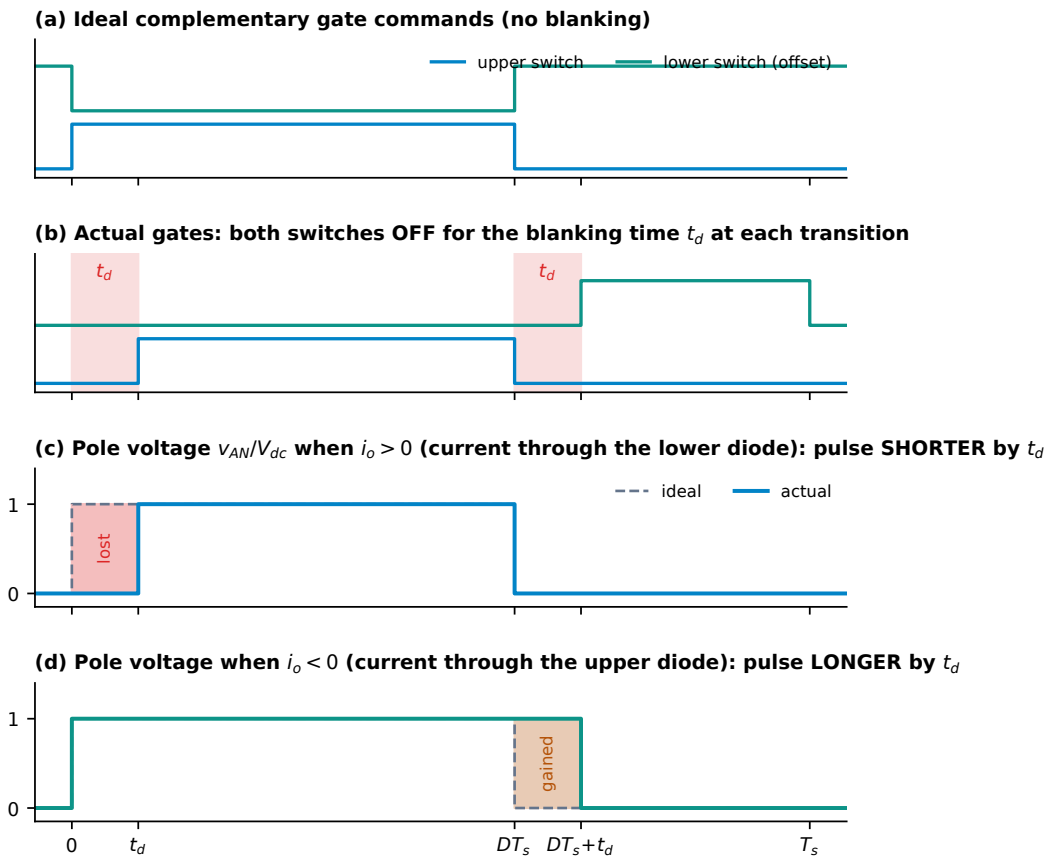


Figure 9: Effect of the blanking time t_d on one inverter leg (computed): (a) ideal complementary gate commands; (b) actual gates, both switches OFF for t_d at every transition; (c) pole voltage for $i_o > 0$: the pulse is shortened by t_d; (d) pole voltage for $i_o < 0$: the pulse is lengthened by t_d. Compare Mohan Fig. 8-31.

turning off (turn-off is slower than turn-on, $t_{off} > t_{on}$). This **shoot-through** is a near short circuit across the DC bus, limited only by parasitic resistance and device drops:

$$I_{\text{fault}} \approx \frac{V_{dc}}{R_{\text{parasitic}}} \rightarrow \text{thousands of amperes} \implies \text{device and DC-link destruction} \tag{47}$$

Implementation of Blanking Time (t_d or t_Δ)

To prevent shoot-through a deliberate delay, the **blanking time** (dead time) t_d, is inserted between the turn-OFF command of one switch and the turn-ON command of the complementary switch of the same leg. The incoming device must not start conducting before the outgoing one has fully turned off:

$$t_d \;>\; t_{off,\text{max}} - t_{d(on),\text{min}} + \Delta t_{\text{driver}} \tag{48}$$

where $t_{d(on),\text{min}}$ is the minimum turn-on delay of the incoming device and Δt_{driver} the spread of gate-driver propagation delays. Typical values: 1–3 μs for IGBTs; 100–500 ns for silicon MOSFETs; < 50 ns for GaN HEMTs.

Adverse Effects and Voltage Error Caused by Blanking Time

During t_d both controlled switches of the leg are OFF and the pole voltage is fixed entirely by the **direction of the load current** through the feedback diodes (Fig. 9):

- **When $i_o > 0$:** the current flows through the lower diode D_4, so the pole voltage is 0 during the dead time even though S_1 is commanded ON. The positive pulse is **shorter by** t_d: area $V_{dc}t_d$ is lost.
- **When $i_o < 0$:** the current flows through the upper diode D_1 and the pole voltage is $+V_{dc}$ during the dead time. The positive pulse is **longer by** t_d.
- The error depends only on the *sign* of the current, not on its magnitude. Averaged over one switching period T_s the pole-voltage error is (Mohan Eq. 8-75):

Average Pole-Voltage Error Due to Blanking Time

$$\Delta V_{Ao} = \frac{t_d}{T_s} V_{dc}\ \text{sgn}(i_A) = f_s\, t_d\, V_{dc}\ \text{sgn}(i_A) \qquad (\text{a drop when } i_A > 0) \tag{49}$$

In the single-phase full bridge with bipolar switching, leg B carries $-i_o$ and its error has the opposite sign in v_{Bo}, so the two legs add: $\Delta V_o = 2 f_s t_d V_{dc}\,\text{sgn}(i_o)$ (Mohan Eq. 8-77). The error is therefore a **square wave in phase with the load current**: its amplitude is $f_s t_d V_{dc}$ per leg and its fundamental component has peak $\frac{4}{\pi} f_s t_d V_{dc}$ (the factor $\frac{4}{\pi}$ comes from the Fourier series of the square wave, not from the averaging).

- **Consequences:**
  1. The fundamental output voltage is reduced, most noticeably at low modulation index (the error is a fixed voltage, independent of m_a).
  2. Low-order harmonics appear: 3rd, 5th, 7th in a single-phase inverter; 5th, 7th, 11th, 13th in a three-phase inverter. They cause torque pulsation in AC motors.
  3. The output voltage is distorted around the current zero crossings.
- **Remedy:** dead-time compensation, i.e. detect the sign of the phase current and add $\pm f_s t_d V_{dc}$ to the voltage command, and use fast devices so that a small t_d is sufficient.

Three-Phase Pulse Width Modulated Voltage Source Inverters

Three-Phase Bridge Inverter Topology and Conduction Mechanics

The three-phase voltage source inverter (Figure 10) is the universal industry-standard power conversion topology for variable-frequency AC motor drives (VFDs), grid-tied solar photovoltaic central inverters, wind turbine grid converters, and electric vehicle traction drives (50 kW $-$ 300 kW).

The topology comprises six controlled switches (S_1 through S_6) arranged in three inverter legs (A, B, C) across a common DC link voltage V_{dc}, with six anti-parallel freewheeling diodes (D_1 through D_6).

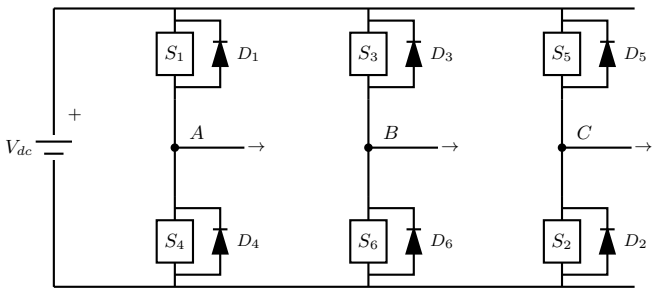


Figure 10: Three-phase voltage source inverter: three legs A, B, C (upper switches S_1, S_3, S_5; lower switches S_4, S_6, S_2), each with an anti-parallel diode, across the DC link V_{dc}. The load may be star or delta connected. (Redrawn; compare Mohan Fig. 8-21.)

Switch Numbering and Firing Sequence

The switches are traditionally numbered according to their natural firing sequence in a balanced three-phase cycle:

- Upper switches: S_1 (Phase A), S_3 (Phase B), S_5 (Phase C).
- Lower switches: S_4 (Phase A), S_6 (Phase B), S_2 (Phase C).
- Firing sequence: $S_1 \to S_2 \to S_3 \to S_4 \to S_5 \to S_6$, spaced by intervals of 60° ($\pi/3$).

Sinusoidal PWM in Three-Phase VSIs

In three-phase SPWM, three balanced sinusoidal reference voltages displaced by 120° are compared against a common high-frequency triangular carrier wave v_{tri}:

$$\begin{aligned} v_{\text{control},A} &= \hat{V}_{\text{control}} \sin(\omega t) \\ v_{\text{control},B} &= \hat{V}_{\text{control}} \sin(\omega t - 120^\circ) \\ v_{\text{control},C} &= \hat{V}_{\text{control}} \sin(\omega t - 240^\circ) = \hat{V}_{\text{control}} \sin(\omega t + 120^\circ) \end{aligned} \tag{50}$$

Pole Voltages and Line-to-Line Voltages

Each leg operates identically to a single-phase half-bridge pole relative to the fictitious DC-bus midpoint o:

$$v_{Ao}(t) = \begin{cases} +V_{dc}/2, & \text{if } v_{\text{control},A} > v_{\text{tri}} \\ -V_{dc}/2, & \text{if } v_{\text{control},A} < v_{\text{tri}} \end{cases} \tag{51}$$

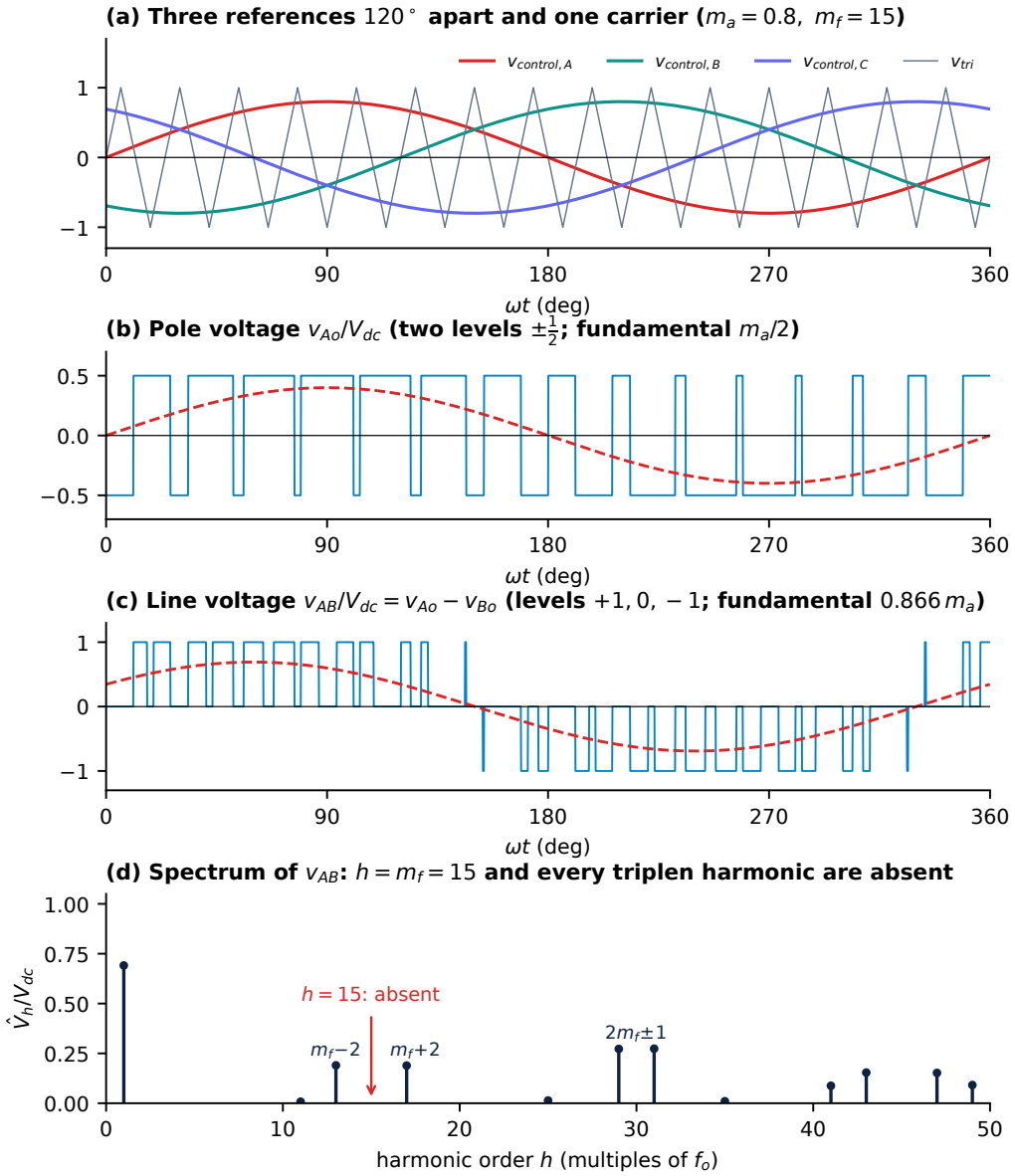


Figure 11: Three-phase SPWM, $m_a = 0.8$, $m_f = 15$ (computed): (a) three references $120°$ apart and one carrier; (b) pole voltage v_{Ao}/V_{dc} (two levels, fundamental $m_a/2$); (c) line voltage $v_{AB}/V_{dc} = v_{Ao} - v_{Bo}$ (levels $+1, 0, -1$, fundamental $0.866\,m_a$); (d) spectrum of v_{AB}: the $h = m_f = 15$ component and all triplen harmonics are absent. Compare Mohan Fig. 8-22.

The line-to-line output voltages are obtained by taking the difference between individual leg pole voltages:

$$\begin{aligned} v_{AB}(t) &= v_{Ao}(t) - v_{Bo}(t) \\ v_{BC}(t) &= v_{Bo}(t) - v_{Co}(t) \\ v_{CA}(t) &= v_{Co}(t) - v_{Ao}(t) \end{aligned} \tag{52}$$

Cancellation of Triplen Harmonics in Three-Phase PWM

A fundamental property of three-phase balanced systems is that **all triplen harmonics** ($n = 3, 9, 15, 21, \dots$) **are identical in magnitude and in-phase across all three phase legs**:

$$v_{Ao,3}(t) = v_{Bo,3}(t) = v_{Co,3}(t) \tag{53}$$

Taking the line-to-line difference completely cancels all triplen harmonics:

$$v_{AB,3}(t) = v_{Ao,3}(t) - v_{Bo,3}(t) \equiv 0! \tag{54}$$

Fundamental Voltage Relationships in 3-Phase SPWM

In the linear modulation range ($0 \leq m_a \leq 1.0$):

1. **Peak Fundamental Pole-to-Midpoint Voltage:**

$$\hat{V}_{Ao1} = m_a \frac{V_{dc}}{2} \tag{55}$$

2. **Peak Fundamental Line-to-Line Voltage:**

$$\hat{V}_{LL1} = \sqrt{3}\hat{V}_{Ao1} = \sqrt{3} m_a \frac{V_{dc}}{2} = \frac{\sqrt{3}}{2} m_a V_{dc} \approx 0.866 m_a V_{dc} \tag{56}$$

3. **RMS Fundamental Line-to-Line Voltage:**

$$V_{LL1(rms)} = \frac{\hat{V}_{LL1}}{\sqrt{2}} = \frac{\sqrt{3}}{2\sqrt{2}} m_a V_{dc} = \frac{\sqrt{6}}{4} m_a V_{dc} \approx 0.6124 m_a V_{dc} \tag{57}$$

4. **Maximum Linear Line-to-Line RMS Voltage ($m_a = 1.0$):**

$$V_{LL1(rms),\text{max}} = 0.6124 V_{dc} \tag{58}$$

Three-Phase Frequency Modulation Ratio (m_f) Selection Criteria

To ensure optimum harmonic elimination in three-phase inverters, m_f must adhere to two strict design criteria:

1. m_f must be an **odd integer**: Ensures half-wave symmetry, completely eliminating all even harmonics.

2. m_f must be an **odd multiple of 3** ($m_f = 9, 15, 21, 27, 33, \dots$):

   - The strongest carrier harmonic of every leg is the one at $h = m_f$ itself. The legs are displaced by $120°$ at the fundamental, so this harmonic is displaced by $120° \times m_f$, which is a multiple of $360°$ when m_f is a multiple of 3: it is then **identical in all three pole voltages**.

- Hence the m_f harmonic (and its odd multiples $3m_f, 5m_f, \dots$) is **cancelled in the line-to-line voltage**.
- The lowest carrier-related harmonics left in v_{AB} are the sidebands $(m_f \pm 2)$, followed by the much smaller $(m_f \pm 4)$ (Fig. 11d).

Overmodulation and Square-Wave Operation in Three-Phase Inverters

Overmodulation Region ($m_a > 1.0$)

When m_a exceeds 1.0 the reference peaks exceed the carrier and pulses are dropped. The fundamental line-to-line voltage keeps rising beyond $0.6124\,V_{dc}$ (RMS) towards the six-step value $0.7797\,V_{dc}$, but no longer in proportion to m_a, and low-order non-triplen harmonics (5th, 7th, 11th, 13th) re-appear in v_{AB}.

Square-Wave (Six-Step) Operation

For very large m_a (practically $m_a \gtrsim 3$–4; the limit is reached only asymptotically) the inverter becomes the classical **six-step (180° conduction) inverter**. Each switch conducts for $180°$, the switching instants are $60°$ apart, and at any instant exactly three switches (one per leg) are ON.

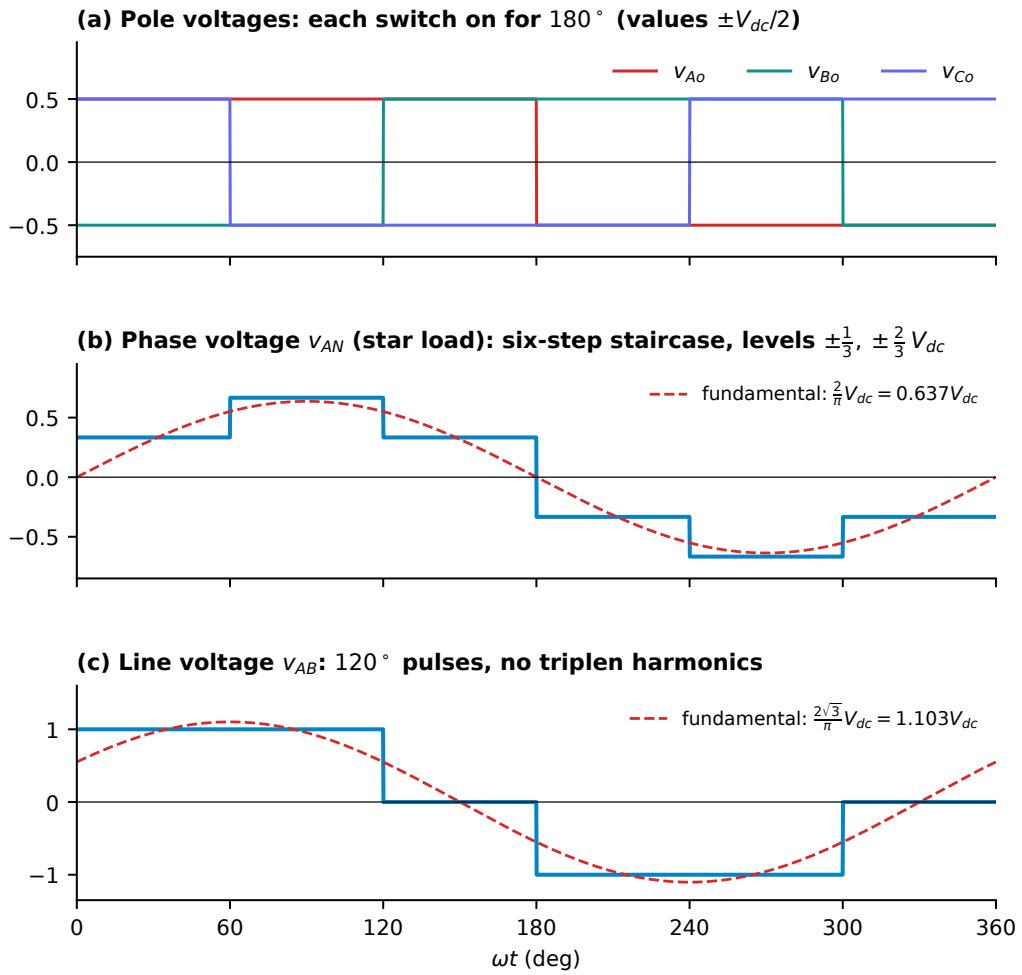


Figure 12: Six-step operation (computed, $\pm V_{dc}/2$ pole voltages): (a) pole voltages displaced by $120°$; (b) phase voltage v_{AN} of a star load, a six-step staircase with levels $\pm\frac{1}{3}V_{dc}$ and $\pm\frac{2}{3}V_{dc}$; (c) line voltage v_{AB}, $120°$ wide pulses separated by $60°$ zero intervals.

- Two legs on the same side connect two load phases in parallel, which then sit in series with the third phase: this is where the levels $\frac{1}{3}$ and $\frac{2}{3}$ come from.

Table 4: Six-step (180° conduction) inverter, star load: conducting switches and voltages in units of V_{dc} (time origin: S_1 turned on at $\omega t = 0$).

Interval	Switches ON	v_{AN}	v_{BN}	v_{CN}	v_{AB}
0°–60°	S_5, S_6, S_1	$+\frac{1}{3}$	$-\frac{2}{3}$	$+\frac{1}{3}$	$+1$
60°–120°	S_6, S_1, S_2	$+\frac{2}{3}$	$-\frac{1}{3}$	$-\frac{1}{3}$	$+1$
120°–180°	S_1, S_2, S_3	$+\frac{1}{3}$	$+\frac{1}{3}$	$-\frac{2}{3}$	0
180°–240°	S_2, S_3, S_4	$-\frac{1}{3}$	$+\frac{2}{3}$	$-\frac{1}{3}$	-1
240°–300°	S_3, S_4, S_5	$-\frac{2}{3}$	$+\frac{1}{3}$	$+\frac{1}{3}$	-1
300°–360°	S_4, S_5, S_6	$-\frac{1}{3}$	$-\frac{1}{3}$	$+\frac{2}{3}$	0

- The line-to-line voltage is a six-step waveform alternating between $+V_{dc}$, 0 and $-V_{dc}$. With the time origin at the centre of the positive pulse of v_{AB}, its Fourier series is:

$$v_{AB}(t) = \sum_{n=1,5,7,11,13,\ldots}^{\infty} \frac{4V_{dc}}{n\pi} \cos\left(\frac{n\pi}{6}\right) \sin(n\omega t) \tag{59}$$

  For all triplen harmonics ($n = 3, 9, 15, \ldots$), $\cos(n\pi/6) = 0$: complete triplen elimination.

- **Peak Fundamental Line-to-Line Voltage:**

$$\hat{V}_{LL1,\text{square}} = \frac{4V_{dc}}{\pi} \cos\left(\frac{\pi}{6}\right) = \frac{2\sqrt{3}}{\pi} V_{dc} \approx 1.1027 V_{dc} \tag{60}$$

- **RMS Fundamental Line-to-Line Voltage:**

$$V_{LL1(rms),\text{square}} = \frac{\hat{V}_{LL1,\text{square}}}{\sqrt{2}} = \frac{\sqrt{6}}{\pi} V_{dc} \approx 0.7797 V_{dc} \tag{61}$$

- **Total RMS Line-to-Line Voltage:**

$$V_{LL(rms)} = \sqrt{\frac{1}{\pi} \int_0^{2\pi/3} V_{dc}^2 \, d(\omega t)} = V_{dc} \sqrt{\frac{2}{3}} \approx 0.8165 V_{dc} \tag{62}$$

- **Phase (line-to-neutral) voltage of a star load:** the staircase v_{AN} has

$$v_{AN}(t) = \sum_{n=1,5,7,11,13,\ldots} \frac{2V_{dc}}{n\pi} \sin(n\omega t)$$

$$\hat{V}_{AN1} = \frac{2V_{dc}}{\pi} \approx 0.637 V_{dc}, \quad V_{AN1(rms)} \approx 0.450 V_{dc}, \quad V_{AN(rms)} = \frac{\sqrt{2}}{3} V_{dc} \approx 0.471 V_{dc}$$

  consistent with $V_{LL} = \sqrt{3}\, V_{AN}$ ($0.8165 = \sqrt{3} \times 0.4714$).

- **Distortion:** $\text{THD}_v = \sqrt{(0.8165/0.7797)^2 - 1} \approx 31.1\%$ for both line and phase voltages (compare 48.3% for the single-phase square wave).

Third Harmonic Injection PWM (THIPWM)

In standard sinusoidal PWM, the maximum linear fundamental line-to-line voltage is limited to $\hat{V}_{LL1} = \frac{\sqrt{3}}{2} V_{dc} \approx 0.866 V_{dc}$ because the peak of each phase reference sine wave cannot exceed the DC rails ($\pm V_{dc}/2$) without causing overmodulation distortion.

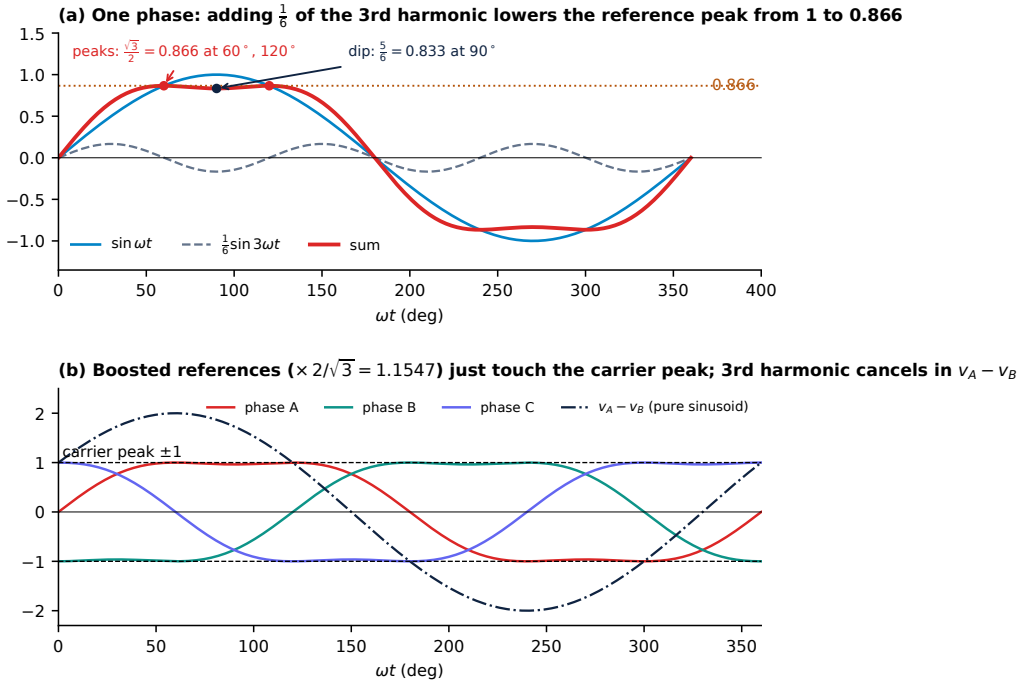


Figure 13: Third-harmonic injection PWM (computed): (a) $\sin\omega t + \frac{1}{6}\sin 3\omega t$ peaks at only 0.866 (at $60°$ and $120°$), with a shallow dip to 0.833 at $90°$; (b) after boosting the fundamental by $2/\sqrt{3}$ the three references just touch the carrier peak, while $v_A - v_B$ remains a pure sinusoid.

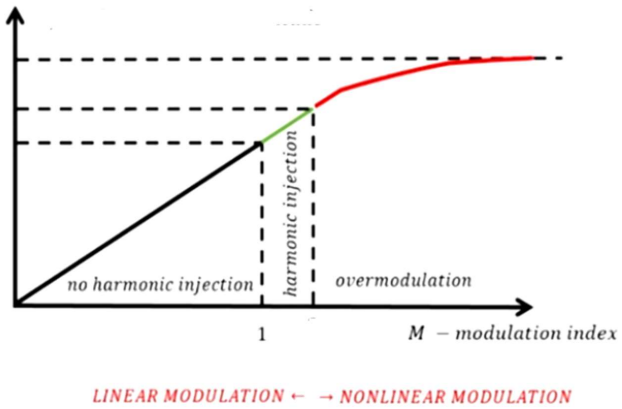


Figure 14: Lecture slide: output versus modulation index M. Harmonic injection extends the linear range beyond $M = 1$ (green) before over-modulation begins (red). (Source: lecture slides)

Mathematical Formulation of THIPWM

A zero-sequence third harmonic, identical in all three phases, is added to each sinusoidal reference:

$$\begin{aligned} v_{\text{control},A}(t) &= \hat{V}_1 \left[\sin(\omega t) + \tfrac{1}{6}\sin(3\omega t)\right] \\ v_{\text{control},B}(t) &= \hat{V}_1 \left[\sin(\omega t - 120^\circ) + \tfrac{1}{6}\sin(3\omega t)\right] \\ v_{\text{control},C}(t) &= \hat{V}_1 \left[\sin(\omega t + 120^\circ) + \tfrac{1}{6}\sin(3\omega t)\right] \end{aligned} \tag{63}$$

(the third harmonic of $\sin(\omega t \mp 120^\circ)$ is $\sin(3\omega t \mp 360^\circ) = \sin 3\omega t$ in every phase). The injected amplitude $\frac{1}{6} \approx 16.7\%$ of the fundamental is the value that gives the largest linear range.

Why Third Harmonic Injection Works

Physics of Third Harmonic Injection

1. **The peak of the composite reference is lower than the fundamental's.** Setting the derivative of $f(\theta) = \sin\theta + \frac{1}{6}\sin 3\theta$ to zero and using $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$:

$$\begin{aligned} f'(\theta) = \cos\theta + \tfrac{1}{2}\cos 3\theta = \tfrac{1}{2}\cos\theta\,\left(4\cos^2\theta - 1\right) = 0 \\ \Longrightarrow\ \cos\theta = 0 \text{ or } \cos\theta = \pm\tfrac{1}{2} \end{aligned}$$

   At $\theta = 60^\circ$ and 120° the maxima are $f = \frac{\sqrt{3}}{2} \approx 0.866$; at $\theta = 90^\circ$ there is only a shallow dip, $f = 1 - \frac{1}{6} = \frac{5}{6} \approx 0.833$. So the peak is flattened to 0.866 of the fundamental amplitude.

2. **15.5% voltage boost in the linear range.** The fundamental can therefore be raised by $\frac{2}{\sqrt{3}} \approx 1.1547$ before the composite reference touches the carrier peak, i.e. m_a may go up to 1.1547 without over-modulation:

$$\begin{aligned} \hat{V}_{LL1,\text{max (THIPWM)}} &= \frac{\sqrt{3}}{2}\,(1.1547)\,V_{dc} = 1.000\,V_{dc} \\ V_{LL1(rms),\text{max (THIPWM)}} &= \frac{V_{dc}}{\sqrt{2}} \approx 0.707\,V_{dc} \end{aligned}$$

   compared with $0.612\,V_{dc}$ for ordinary SPWM: a gain of 15.5% without leaving the linear region.

3. **No distortion at the load.** The injected term is the same in all three phases (zero-sequence), so it cancels in every line-to-line voltage:

$$v_{AB}(t) = v_{Ao}(t) - v_{Bo}(t) = \text{pure fundamental (+ switching harmonics)} \tag{64}$$

   and, with an isolated star point, it drives no load current.

4. **Same DC-bus utilisation as space-vector PWM.** Space-vector modulation reaches the same maximum linear output (1.1547 in m_a terms); it effectively injects a triangular (min–max) zero-sequence signal instead of a sinusoidal third harmonic, but THIPWM achieves the same range with simple carrier comparators.

Table 5: Maximum fundamental line-to-line RMS voltage of a three-phase inverter, as a fraction of V_{dc}.

Modulation	$V_{LL1(rms),\max}/V_{dc}$	Relative to SPWM
SPWM, linear range ($m_a = 1$)	0.612	1.000
THIPWM / SVPWM, linear range	0.707	1.155
Six-step (square wave)	0.780	1.273

Current Source Inverters (CSI) and Detailed Comparison with VSI

Operating Principle of Single-Phase Current Source Inverters (CSI)

In contrast to a Voltage Source Inverter (which is fed from a stiff, low-impedance DC voltage source), a **Current Source Inverter (CSI)** is fed from an adjustable direct current source possessing **high internal impedance**. In practice, this is realized by placing a large smoothing choke (L_{dc}) in series with a controlled DC power supply or phase-controlled thyristor rectifier.

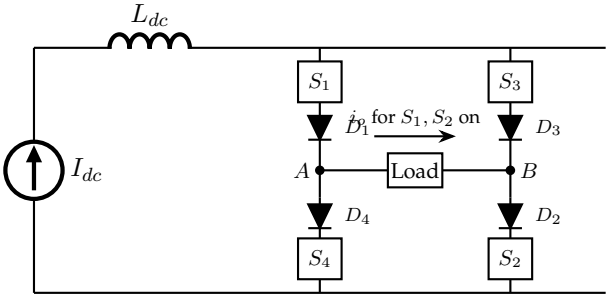


Figure 15: Single-phase IGBT-based Current Source Inverter (CSI) with series blocking diodes and massive DC smoothing inductor L_{dc}.

Circuit Characteristics of the CSI

1. **Constant Input Current (I_{dc}):** Due to the large inductance L_{dc}, the DC link current cannot change instantaneously:
$$i_{dc}(t) \approx I_{dc} = \text{constant (ripple-free)} \tag{65}$$

2. **Square-Wave Output Current Waveform:** Gating switch pair (S_1, S_2) directs $+I_{dc}$ through the load, while gating (S_3, S_4) directs $-I_{dc}$ through the load. The output current $i_o(t)$ is therefore a **clean square wave alternating between $+I_{dc}$ and $-I_{dc}$.**

3. **Need for Series Blocking Diodes ($D_1 - D_4$):**
   - Standard power MOSFETs and IGBTs have anti-parallel body diodes or low reverse-blocking voltage ratings (< 20 V).
   - In a CSI, the load voltage is alternating AC, meaning the switches must withstand **large reverse voltages**.
   - Fast-recovery diodes are placed strictly in **series** with each controlled switch to provide the necessary reverse voltage blocking capability!

4. **Inherent Short-Circuit Immunity:** In a VSI, accidental simultaneous conduction of upper and lower switches causes an explosive shoot-through short circuit. In sharp contrast, in a CSI, **simultaneous conduction of upper and lower switches merely circulates the constant DC current I_{dc} without any current spike!**

5. **Open-Circuit Danger:** In a CSI, an open circuit in the load or DC path is catastrophic ($L\frac{di}{dt} \to \infty$, generating thousands of volts of inductive overvoltage arc breakdown). An overlap period is mandatory during commutation!

Output Current of the Single-Phase CSI (Dual of the VSI Voltage)

With ideal switches and a constant I_{dc}, the output current is a square wave of amplitude $\pm I_{dc}$, the exact dual of the VSI output voltage:

$$i_o(t) = \sum_{n=1,3,5,\ldots} \frac{4I_{dc}}{n\pi} \sin(n\omega t)$$

$$\hat{I}_{o1} = \frac{4I_{dc}}{\pi} \approx 1.273 I_{dc}, \qquad I_{o1(rms)} = \frac{2\sqrt{2}}{\pi} I_{dc} \approx 0.900 I_{dc}, \qquad \text{THD}_i = 48.3\%$$

The output *voltage* is not imposed by the switches but developed by this current in the load. A parallel RLC load tuned to f_o (high Q) offers a high impedance only at the fundamental, so the harmonic currents produce almost no voltage and v_o is nearly sinusoidal (Fig. 16).

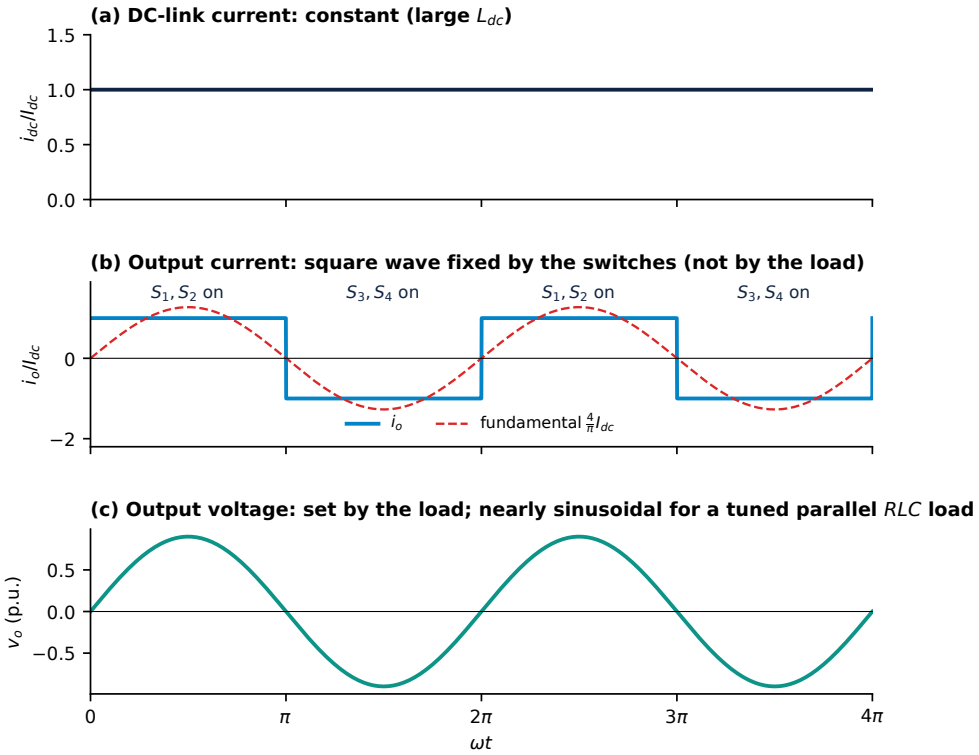


Figure 16: Single-phase CSI waveforms (computed): (a) constant DC-link current; (b) square-wave output current and its fundamental $\frac{4}{\pi} I_{dc}$; (c) near-sinusoidal output voltage across a tuned parallel RLC load.

Comprehensive Technical Comparison: VSI vs. CSI

The comparison between Voltage Source Inverters and Current Source Inverters is a quintessential KTU university examination question (often awarded 6 to 9 marks). Table 6 provides an exhaustive, multi-dimensional comparison.

Table 6: Definitive Technical Comparison: Voltage Source Inverter (VSI) vs. Current Source Inverter (CSI)

Feature / Parameter	Voltage Source Inverter (VSI)	Current Source Inverter (CSI)
DC Source Characteristic	Stiff DC voltage source with very small internal impedance	Stiff DC current source with very high internal impedance
Energy Storage Element	Large DC-link capacitor bank (C_{dc}) across input terminals	Large DC-link smoothing inductor / choke (L_{dc}) in series
Output Voltage Waveform	Controlled square wave or PWM pulsed waveform (independent of load)	Determined entirely by load impedance and current
Output Current Waveform	Determined entirely by the connected load impedance	Pure square wave or PWM current waveform (independent of load)
Freewheeling / Blocking Diodes	Requires freewheeling diodes in **anti-parallel** with switches	Requires blocking diodes in **series** with switches
Regeneration Capability	Requires a four-quadrant dual converter or active front-end rectifier	Natural and simple: just reverse the DC voltage polarity while I_{dc} remains unidirectional
Short-Circuit Behavior	Highly vulnerable: leg shoot-through creates catastrophic short circuit	Inherent short-circuit ride-through; fault current limited by I_{dc}
Open-Circuit Behavior	Safe: open circuit causes no damage	Extremely hazardous: open-circuit produces destructive inductive $L\frac{di}{dt}$ voltage spikes
Switching Overlap Requirements	Strict **blanking time (dead-time)** mandatory to avoid shoot-through	Strict **overlap time** mandatory to avoid open-circuiting the DC choke
Switch Ratings	High current rating, unidirectional voltage blocking	High reverse voltage blocking capability required
Dynamic Response	Extremely fast (voltage switched instantaneously)	Slower due to large magnetic time constant of L_{dc}
Size, Weight & Cost	Compact and lightweight (uses electrolytic/film capacitors)	Bulky, heavy, and expensive due to massive iron-core DC reactor
Primary Industrial Use	General-purpose industrial VFDs, solar inverters, EV traction, UPS	High-power multi-megawatt synchronous motor drives, induction heating

Utility Interface, Harmonics, IEEE 519 Standard, and Active PFC

Degradation of the Electric Utility Interface

Conventional line-frequency diode rectifiers and phase-controlled thyristor converters feeding DC filter capacitors draw **highly non-sinusoidal, pulsed input currents** from the electric utility AC distribution grid.

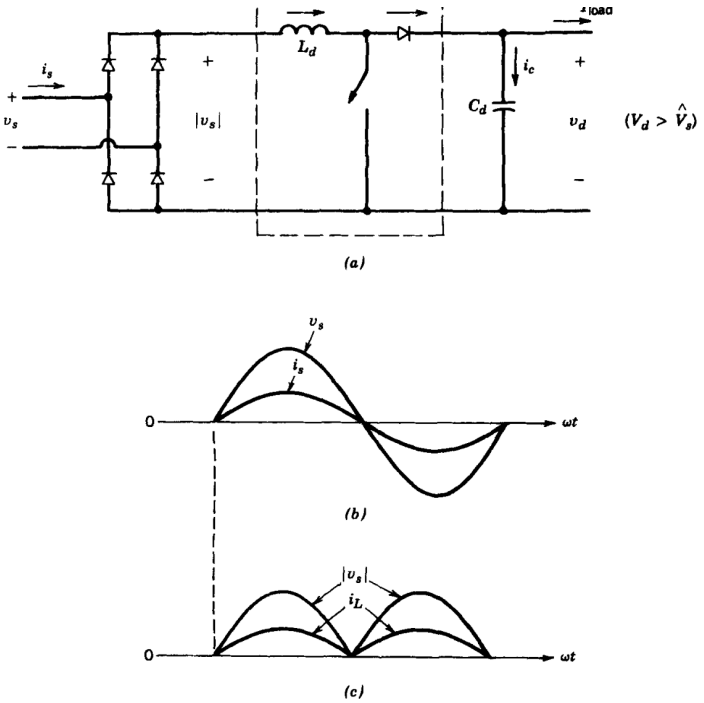


Figure 18-4 Active harmonic filtering: (*a*) step-up converter for current shaping; (*b*) line waveforms; (*c*) v_s and i_L.

Figure 17: Optimizing the utility interface: (a) step-up (boost) converter used for current shaping; (b) line waveforms v_s and i_s; (c) rectified voltage $|v_s|$ and inductor current i_L, which follows $|v_s|$. (Source: Ned Mohan, Fig. 18-4)

Harmonic Generation and Detrimental Impacts on Power Systems

1. **Excessive Neutral Current in Three-Phase Four-Wire Systems:** Triplen current harmonics (3rd, 9th, 15th, ...) generated by single-phase computer SMPS and electronic ballasts do not cancel in the neutral conductor; they add arithmetically in-phase:

$$I_N = \sqrt{I_{N,3}^2 + I_{N,9}^2 + \dots} \approx 3 \cdot I_{\text{phase},3} \tag{66}$$

   This causes neutral conductors to carry up to $1.73 - 2.0\times$ the phase current, leading to transformer overheating and neutral cable fires!

2. **Transformer and Generator De-rating:** High-frequency harmonic currents induce massive eddy current ($P_{\text{eddy}} \propto f^2$) and stray load losses in magnetic cores, requiring costly K-factor rated transformers.

3. **Voltage Distortion Across the Power Grid:** The harmonic current I_h drawn through the finite utility grid source impedance ($Z_s = R_s + j\omega L_s$) causes voltage drops that distort the supply voltage at the Point of Common Coupling (PCC):

$$V_{h,\text{PCC}} = I_h \cdot Z_{s,h} \tag{67}$$

   This degrades the voltage waveform delivered to neighboring consumers.

4. **Harmonic Resonance with Power Factor Correction Capacitors:** Grid inductances and utility shunt capacitor banks form parallel resonant circuits ($f_{\text{res}} = \frac{1}{2\pi\sqrt{L_{\text{grid}}C_{\text{pfc}}}}$). Harmonic currents near f_{res} excite catastrophic overvoltage resonances, exploding capacitor banks!

Power Factor in Non-Sinusoidal Circuits

When the supply voltage is purely sinusoidal $v_s(t) = \sqrt{2}V_s \sin(\omega t)$ but the input current is distorted by non-linear electronics:

$$i_s(t) = \sqrt{2}I_{s1}\sin(\omega t - \phi_1) + \sum_{h=2}^{\infty}\sqrt{2}I_{sh}\sin(h\omega t - \phi_h) \tag{68}$$

The total RMS current is given by:

$$I_s = \sqrt{I_{s1}^2 + \sum_{h=2}^{\infty} I_{sh}^2} = \sqrt{I_{s1}^2 + I_{\text{dist}}^2} \tag{69}$$

Active Real Power (P) and Apparent Power (S)

Because the cross-product of orthogonal sinusoids of different frequencies integrates to zero over a full cycle, **only the fundamental current component I_{s1} produces average real power**:

$$P = \frac{1}{T}\int_0^T v_s(t)i_s(t)dt = V_s I_{s1}\cos\phi_1 \tag{70}$$

The total apparent power is:

$$S = V_s I_s = V_s\sqrt{I_{s1}^2 + I_{\text{dist}}^2} \tag{71}$$

Displacement Power Factor (DPF) vs. Distortion Power Factor

The true Power Factor (PF) is defined as the ratio of real active power to total apparent power:

$$\text{PF} = \frac{P}{S} = \frac{V_s I_{s1}\cos\phi_1}{V_s I_s} = \left(\frac{I_{s1}}{I_s}\right)\cdot\cos\phi_1 \tag{72}$$

True Power Factor Decomposition

$$\text{PF} = K_{\text{dist}}\cdot\text{DPF} = \left(\frac{I_{s1}}{I_s}\right)\cos\phi_1 \tag{73}$$

where:

- $\text{DPF} = \cos\phi_1$ is the **Displacement Power Factor** (phase displacement of fundamental current relative to voltage).
- $K_{\text{dist}} = \frac{I_{s1}}{I_s}$ is the **Distortion Factor** (purity of the current waveform).

Relationship Between Distortion Factor and Total Harmonic Distortion (THD)

The Total Harmonic Distortion of the current is defined as:

$$\text{THD}_i = \frac{\sqrt{\sum_{h=2}^{\infty} I_{sh}^2}}{I_{s1}} = \frac{\sqrt{I_s^2 - I_{s1}^2}}{I_{s1}} = \sqrt{\left(\frac{I_s}{I_{s1}}\right)^2 - 1} \tag{74}$$

Rearranging gives the direct link between Distortion Factor and THD_i:

$$\frac{I_{s1}}{I_s} = \frac{1}{\sqrt{1 + \text{THD}_i^2}} \tag{75}$$

Substituting into equation (73) yields the foundational formula:

Master Formula: True Power Factor from THD

$$\text{PF} = \frac{\cos\phi_1}{\sqrt{1 + \text{THD}_i^2}} \tag{76}$$

Crucial Exam Insight

Even if the displacement power factor is unity ($\cos\phi_1 = 1.0$), a conventional bridge rectifier with a typical $\text{THD}_i = 80\%$ has a true power factor of only:

$$\text{PF} = \frac{1.0}{\sqrt{1 + (0.80)^2}} = \frac{1.0}{\sqrt{1 + 0.64}} = \frac{1.0}{1.2806} = 0.7808 \implies 78.1\%! \tag{77}$$

This proves that **harmonic distortion alone degrades power factor**, regardless of capacitive/inductive phase shifts!

IEEE 519 Standard for Harmonic Control in Electric Power Systems

The **IEEE 519 Standard** (*IEEE Standard for Harmonic Control in Electric Power Systems*; current edition IEEE Std 519-2022, which replaced the 2014 edition) sets limits on harmonic voltage and current distortion at the **Point of Common Coupling (PCC)**, the point on the supply side nearest the customer at which other customers may be connected. The customer is responsible for the *current* limits and the utility for the *voltage* limits. The limits become binding when the utility adopts them in its interconnection agreement.

PCC and Total Demand Distortion (TDD)

IEEE 519 evaluates current distortion using **Total Demand Distortion (TDD)** rather than conventional THD:

$$\text{TDD} = \frac{\sqrt{\sum_{h=2}^{\infty} I_h^2}}{I_L} \tag{78}$$

where I_L is the **maximum demand load current** (fundamental component at the PCC under normal operating conditions; typically the average of the monthly maximum demands over the previous 12 months, each measured over a 15- or 30-minute demand interval), while THD is calculated relative to instantaneous fundamental current I_1. Under light load, THD can be artificially astronomical ($> 100\%$) while the actual harmonic Amperes injected into the grid are tiny; TDD prevents this distortion artifact!

IEEE 519 Current Distortion Limits (for Systems Rated 120 V to 69 kV)

The permissible harmonic current limits depend directly on the **Short-Circuit Ratio (SCR)** at the PCC:

$$\text{SCR} = \frac{I_{sc}}{I_L} \tag{79}$$

where I_{sc} is the maximum short-circuit current available at the PCC, and I_L is the maximum fundamental load demand current. A stiffer grid (higher I_{sc}) can absorb more harmonic current without suffering severe voltage distortion.

Table 7: IEEE 519 Current Distortion Limits at the PCC (120 V – 69 kV)

I_{sc}/I_L **Ratio**	$h < 11$	$11 \leq h < 17$	$17 \leq h < 23$	$23 \leq h < 35$	$35 \leq h \leq 50$	**Max TDD (%)**
< 20*	4.0%	2.0%	1.5%	0.6%	0.3%	**5.0**%
$20 < 50$	7.0%	3.5%	2.5%	1.0%	0.5%	**8.0**%
$50 < 100$	10.0%	4.5%	4.0%	1.5%	0.7%	**12.0**%
$100 < 1000$	12.0%	5.5%	5.0%	2.0%	1.0%	**15.0**%
> 1000	15.0%	7.0%	6.0%	2.5%	1.4%	**20.0**%

*All power generation equipment is restricted to these lowest values regardless of actual I_{sc}/I_L.
Odd harmonics shown. Even harmonics: limited to 25% of these values in the 2014 edition; in IEEE 519-2022 limited to 50% for $h \leq 6$ and equal to the odd limits for $h > 6$. Current distortion that results in a DC offset (e.g. half-wave converters) is not allowed.

Table 8: IEEE 519 Voltage Distortion Limits at the PCC

Bus Voltage at PCC (V)	**Individual Harmonic (%)**	**Total Voltage Distortion THD**$_v$ **(%)**
$V \leq 1.0$ kV	5.0%	**8.0**%
1.0 kV $< V \leq$ 69 kV	3.0%	**5.0**%
69 kV $< V \leq$ 161 kV	1.5%	**2.5**%
$V >$ 161 kV	1.0%	**1.5**%

Active Shaping of the Input Line Current (Power Factor Correction)

To meet the rigorous harmonic constraints of IEEE 519 and IEC 61000-3-2, modern power supplies employ an **Active Power Factor Correction (PFC) Boost Converter** operating at high switching frequency ($50 - 200$ kHz) directly downstream of the uncontrolled diode bridge.

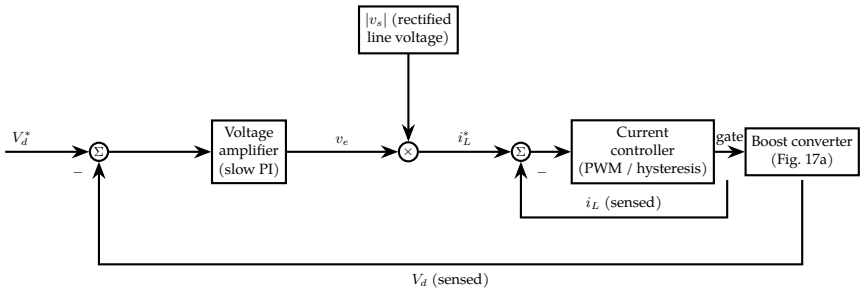


Figure 18: Control architecture of the active PFC boost converter: slow outer voltage loop (regulates V_d, produces v_e), multiplier forming the reference $i_L^* = K v_e |v_s|$, and fast inner current loop driving the switch. (Redrawn; compare Mohan Fig. 18-5.)

Circuit Configuration and Dual-Loop Control Mechanics

As depicted in Figures 17 and 18, the active PFC step-up converter consists of a full-wave diode bridge, boost inductor L, power MOSFET/IGBT switch S, fast diode D, and DC bulk capacitor C_d.

The control system employs two cascaded feedback control loops:

1. **Slow Outer Voltage Loop (DC Bus Regulation):**
   - Senses the DC link voltage V_d and compares it to a precise reference V_d^* (typically 390 V – 400 V DC for universal AC mains of 85 V – 265 V).
   - A proportional-integral (PI) voltage controller generates an error voltage signal v_e.
   - **Deliberate Low Bandwidth:** The bandwidth of this loop is intentionally set very low (10 – 20 Hz) to prevent the controller from responding to the inevitable 100 Hz or 120 Hz ripple present on V_d.

2. **Analog Multiplier Stage:**
   - Multiplies the slowly varying error signal v_e with a rectified voltage template signal $|v_s(t)|$ obtained directly from a potential divider on the AC mains:
   $$i_L^*(t) = K \cdot v_e \cdot |v_s(t)| = \hat{I}_L^* |\sin(\omega t)| \tag{80}$$
   This establishes an instantaneous reference current $i_L^*(t)$ that is guaranteed to have the exact same shape and phase as the AC mains voltage!
3. **Fast Inner Current Loop (Waveform Shaping):**
   - Senses the instantaneous inductor current $i_L(t)$ using a shunt resistor or Hall-effect sensor.
   - A high-bandwidth current controller (Average Current Mode Control or Peak Current Control) forces $i_L(t)$ to track $i_L^*(t)$ cycle-by-cycle at switching frequencies of $50 - 200$ kHz.
   - When switch S is closed: Inductor current rises ($di_L/dt = |v_s|/L$).
   - When switch S is open: Inductor current discharges into the load ($di_L/dt = (|v_s| - V_d)/L < 0$).

Key Operating Metrics and Benefits of Active PFC

- **Near-Unity Power Factor:** The input line current $i_s(t)$ is a clean, continuous sinusoid precisely in-phase with the line voltage ($\phi_1 \approx 0 \implies$ DPF ≈ 1.0).
- **Ultra-Low Harmonic Distortion:** THD$_i$ is suppressed from $> 80\%$ down to $< 3\% - 5\%$, easily satisfying IEEE 519 limits under all operating tiers!
- **Universal Input Voltage Range:** Active boost PFC automatically accommodates universal AC line inputs (85 V$_{\text{rms}} - 265$ V$_{\text{rms}}$, $47 - 63$ Hz) while maintaining a rock-steady regulated 400 V DC bus for downstream inverters or DC-DC stages!

Ten Comprehensive University Exam Solved Numerical Problems

Problem 1: Single-Phase Full-Wave AC Voltage Controller with Resistive Load

Problem 1: ACVC with R Load Design & Analysis

Statement: A single-phase full-wave AC voltage controller is supplied from a 230 V, 50 Hz single-phase source and feeds a purely resistive load of $R = 10\,\Omega$. The firing delay angle is set to $\alpha = \pi/3$ ($60°$). Determine:

1. The RMS output voltage V_{or}.
2. The RMS load current I_{or} and real active power dissipated in the load P_o.
3. The input power factor PF.
4. The average and RMS current ratings required for each individual thyristor.

Solution:

1. **RMS Output Voltage (V_{or}):** Using the master analytical formula for resistive load (equation (6)):
$$V_{or} = V_s \sqrt{1 - \frac{\alpha}{\pi} + \frac{\sin(2\alpha)}{2\pi}}$$

Here, $\alpha = \frac{\pi}{3} = 60^\circ$ (1.0472 rad), and $2\alpha = \frac{2\pi}{3} = 120^\circ$.

$$\sin(2\alpha) = \sin(120^\circ) = \frac{\sqrt{3}}{2} \approx 0.8660$$

Substituting the numerical values into the square root expression:

$$\text{Term} = 1 - \frac{\pi/3}{\pi} + \frac{0.8660}{2\pi} = 1 - \frac{1}{3} + \frac{0.8660}{6.2832} = \frac{2}{3} + 0.1378 = 0.6667 + 0.1378 = 0.8045$$

$\sqrt{0.8045} \approx 0.8969$

Therefore, the RMS load voltage is:

$$\mathbf{V_{or}} = 230 \times 0.8969 = \mathbf{206.3}\ \text{V}$$

2. **RMS Load Current and Active Load Power:**

$$\mathbf{I_{or}} = \frac{V_{or}}{R} = \frac{206.3\ \text{V}}{10\,\Omega} = \mathbf{20.63}\ \text{A}$$

The real power absorbed by the load resistor is:

$$\mathbf{P_o} = I_{or}^2 \cdot R = (20.63)^2 \times 10 = \mathbf{4256}\ \text{W} = \mathbf{4.256}\ \text{kW}$$

3. **Input Power Factor (PF):**

$$\text{PF} = \frac{P_o}{V_s I_{or}} = \frac{V_{or}}{V_s} = \frac{206.3}{230} = \mathbf{0.897}\ \text{(lagging)}$$

4. **Thyristor Current Ratings:** Each thyristor carries current for one half-cycle. Peak supply voltage is $V_m = \sqrt{2} \times 230 = 325.27$ V.

$$\mathbf{I_{T(avg)}} = \frac{V_m}{2\pi R}(1 + \cos\alpha) = \frac{325.27}{2\pi \times 10}(1 + \cos 60^\circ) = \frac{32.527}{6.2832}(1 + 0.5) = 5.177 \times 1.5 = \mathbf{7.77}\ \text{A}$$

$$\mathbf{I_{T(rms)}} = \frac{I_{or}}{\sqrt{2}} = \frac{20.63}{\sqrt{2}} = \mathbf{14.59}\ \text{A}$$

Problem 2: Single-Phase AC Voltage Controller with Inductive RL Load

Problem 2: ACVC with RL Load and Extinction Angle

Statement: A 230 V, 50 Hz single-phase AC voltage controller feeds a series RL load with $R = 4\,\Omega$ and $\omega L = 3\,\Omega$.

1. Determine the minimum permissible firing angle α_{min} for controllable operation.
2. For a firing angle $\alpha = 60^\circ$, find the extinction angle β, the conduction angle γ and the RMS output voltage V_{or}.

Solution:

1. **Minimum Firing Angle ($\alpha_{\min}$):**

$$Z = \sqrt{R^2 + (\omega L)^2} = \sqrt{4^2 + 3^2} = 5.0\,\Omega$$
$$\phi = \arctan\left(\frac{\omega L}{R}\right) = \arctan\left(\frac{3}{4}\right) = 36.87^\circ = 0.6435 \text{ rad}$$

For independent phase control the delay angle must satisfy $\alpha \geq \phi$:

$$\alpha_{\min} = \phi = \mathbf{36.87}^\circ$$

2. **Extinction angle β (solve the transcendental equation).** With $\alpha = 60^\circ > \phi$ the current is discontinuous. β is the first root beyond π of

$$f(\beta) = \sin(\beta - \phi) - \sin(\alpha - \phi)\, e^{-\frac{R}{\omega L}(\beta - \alpha)} = 0$$
$$\sin(\alpha - \phi) = \sin 23.13^\circ = 0.3928, \qquad \frac{R}{\omega L} = 1.3333$$

Evaluate f for two trial values and interpolate:

$$f(216^\circ) = \sin(179.13^\circ) - 0.3928\, e^{-1.3333 \times 2.7227} = 0.01518 - 0.01043 = +0.00476$$
$$f(217^\circ) = \sin(180.13^\circ) - 0.3928\, e^{-1.3333 \times 2.7402} = -0.00227 - 0.01019 = -0.01246$$

$$\beta \approx 216^\circ + \frac{0.00476}{0.00476 + 0.01246} \times 1^\circ \approx \mathbf{216.3}^\circ$$
$$\gamma = \beta - \alpha = \mathbf{156.3}^\circ = 2.728 \text{ rad} \quad (< 180^\circ : \text{ discontinuous conduction})$$

3. **RMS Output Voltage:** Using equation (18):

$$2\beta = 432.6^\circ = 72.6^\circ \implies \sin 2\beta = 0.9540, \qquad 2\alpha = 120^\circ \implies \sin 2\alpha = 0.8660$$
$$\frac{\sin 2\beta - \sin 2\alpha}{2} = \frac{0.9540 - 0.8660}{2} = 0.0440$$
$$\frac{1}{\pi}\left[\gamma - 0.0440\right] = \frac{2.728 - 0.044}{\pi} = 0.8543, \qquad \sqrt{0.8543} = 0.9243$$

$$\mathbf{V_{or}} = 230 \times 0.9243 = \mathbf{212.6}\text{ V}$$

Check: for a pure resistor at the same α the output would be $V_{or} = 230\sqrt{1 - \frac{1}{3} + \frac{\sin 120^\circ}{2\pi}} = 206.3$ V (conduction 120°). The inductor keeps T_1 conducting for 156.3° instead, so the RL result 212.6 V is correctly larger.

Problem 3: Single-Phase Full-Bridge Square-Wave Inverter Analysis

Problem 3: Square-Wave Full-Bridge Inverter Analysis

Statement: A single-phase full-bridge inverter is fed from a 48 V battery bank and delivers power to a series RL load with $R = 5\,\Omega$ and $L = 20$ mH at fundamental frequency $f = 50$ Hz. Determine:

1. The total RMS output voltage V_{or} and the fundamental RMS output voltage $V_{o1(rms)}$.
2. The fundamental load current $i_{o1}(t)$ and its RMS magnitude $I_{o1(rms)}$.
3. The fundamental active power P_1 absorbed by the load.

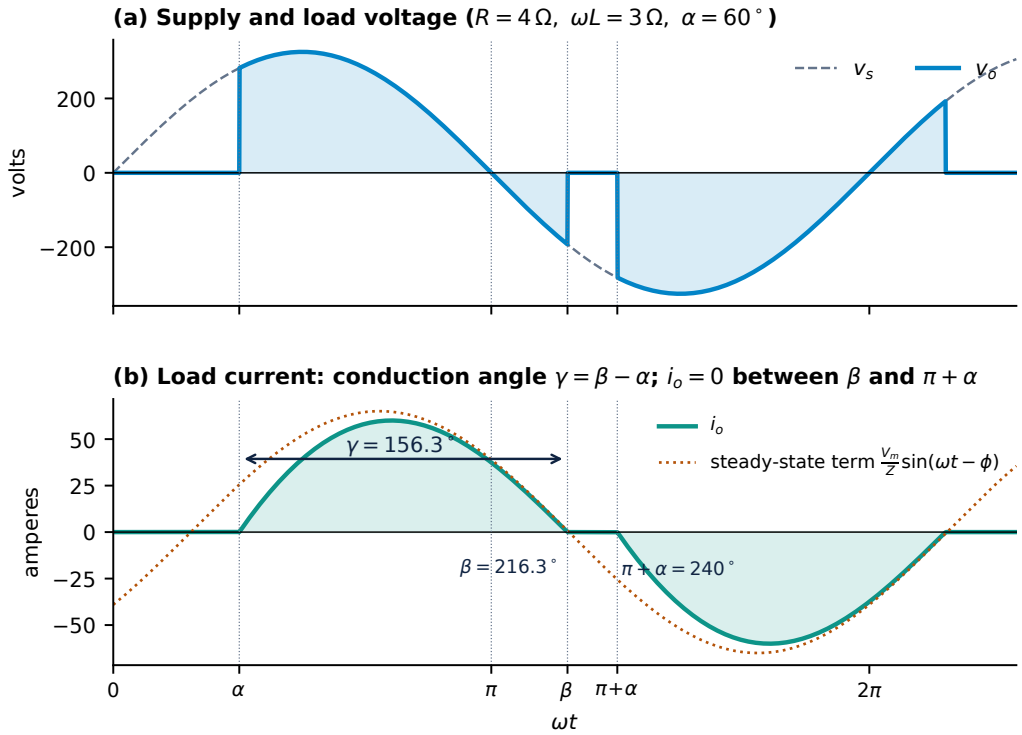


Figure 19: Problem 2 (computed from the exact solution): load voltage and current for $R = 4\,\Omega$, $\omega L = 3\,\Omega$, $\alpha = 60°$. Conduction ends at $\beta = 216.3°$; $i_o = 0$ until T_2 is fired at $\pi + \alpha = 240°$.

4. The total harmonic distortion of the output voltage THD_v.
5. The exact steady-state load current: initial/peak value, zero-crossing instant, and the devices that conduct.

Solution:

1. **Output Voltages:**

$$\mathbf{V_{or}} = V_{dc} = \mathbf{48.0}\ \text{V}$$

$$\mathbf{V_{o1(rms)}} = \frac{2\sqrt{2}}{\pi} V_{dc} = 0.9003 \times 48 = \mathbf{43.21}\ \text{V}, \qquad \hat{V}_{o1} = \frac{4V_{dc}}{\pi} = 61.12\ \text{V}$$

2. **Fundamental Load Current:** $\omega = 2\pi \times 50 = 314.16\ \text{rad/s}$.

$$X_{L1} = \omega L = 314.16 \times 20 \times 10^{-3} = 6.283\,\Omega$$

$$Z_1 = \sqrt{5^2 + 6.283^2} = 8.03\,\Omega, \qquad \phi_1 = \arctan\left(\frac{6.283}{5}\right) = 51.49^\circ$$

$$\hat{I}_{o1} = \frac{61.12}{8.03} = 7.61\ \text{A} \implies \mathbf{I_{o1(rms)}} = \frac{7.61}{\sqrt{2}} = \mathbf{5.38}\ \text{A}, \qquad \mathbf{i_{o1}(t)} = \mathbf{7.61}\sin(\mathbf{314.16t} - \mathbf{51.49}^\circ)\ \text{A}$$

3. **Fundamental Active Power:**

$$\mathbf{P_1} = I_{o1(rms)}^2 R = (5.38)^2 \times 5 \approx \mathbf{144.8}\ \text{W}$$

4. **Total Harmonic Distortion of the voltage:**

$$\text{THD}_\mathbf{v} = \sqrt{\left(\frac{V_{or}}{V_{o1(rms)}}\right)^2 - 1} = \sqrt{\left(\frac{48}{43.21}\right)^2 - 1} = \sqrt{1.2337 - 1} = \mathbf{48.34}\%$$

5. **Exact current (Eq. (35)):** $\tau = L/R = 4$ ms, $T = 20$ ms, $\frac{T}{4\tau} = 1.25$, $\tanh 1.25 = 0.8483$.

$$I_0 = -\frac{48}{5} \times 0.8483 = -8.14 \text{ A}, \qquad I_{\max} = +\mathbf{8.14} \text{ A}$$
$$t_0 = \tau \ln(1 + 0.8483) = 4 \text{ ms} \times 0.614 = 2.46 \text{ ms} \implies \omega t_0 = \mathbf{44.2}^\circ$$

For $0 < \omega t < 44.2^\circ$ the current is negative and flows through D_1, D_2 (energy returned to the battery); from 44.2° to 180° it flows through S_1, S_2; the second half-cycle repeats with D_3, D_4 and S_3, S_4. The exact RMS current is 5.44 A (total load power 148.1 W); the harmonic currents are small (3rd: 1.05 A peak, $\text{THD}_i \approx 15\%$), which is why the fundamental analysis already gives $P \approx P_1$ within 2%.

Problem 4: SPWM Full-Bridge Inverter with Bipolar Voltage Switching

Problem 4: Bipolar SPWM Inverter Parameter Sizing

Statement: A single-phase full-bridge inverter with sinusoidal PWM and bipolar switching is supplied from a $V_{dc} = 300$ V DC rail. The carrier frequency is $f_s = 1.95$ kHz and the output frequency is $f_o = 50$ Hz. The triangular carrier has amplitude $\hat{V}_{\text{tri}} = 5$ V and the control reference has amplitude $\hat{V}_{\text{control}} = 4$ V.

1. Determine m_a and m_f.
2. Calculate the peak and RMS fundamental output voltages.
3. Identify the frequencies and peak amplitudes of the dominant harmonic components.

Solution:

1. **Modulation Indices:**

$$\mathbf{m_a} = \frac{4 \text{ V}}{5 \text{ V}} = \mathbf{0.80}$$
$$\mathbf{m_f} = \frac{f_s}{f_o} = \frac{1950}{50} = \mathbf{39}$$

$m_f = 39$ is an odd integer (synchronous PWM, half-wave symmetry, no even harmonics), as required for bipolar switching.

2. **Fundamental Output Voltages** ($m_a \leq 1$: linear range):

$$\hat{\mathbf{V}}_{\mathbf{o1}} = m_a V_{dc} = 0.80 \times 300 = \mathbf{240.0} \text{ V}$$
$$\mathbf{V_{o1(rms)}} = \frac{240}{\sqrt{2}} = \mathbf{169.7} \text{ V}$$

3. **Dominant Harmonics** (Table 2, column $m_a = 0.8$; peak value = table entry $\times$ 300 V):

Order	Frequency	Table entry	Peak amplitude
$m_f = 39$	1950 Hz	0.818	245.4 V
$m_f \pm 2 = 37,\ 41$	1850 Hz, 2050 Hz	0.220	66.0 V
$2m_f \pm 1 = 77,\ 79$	3850 Hz, 3950 Hz	0.314	94.2 V
$2m_f \pm 3 = 75,\ 81$	3750 Hz, 4050 Hz	0.139	41.7 V

Notice that the carrier-frequency harmonic (245 V) is larger than the 240 V fundamental: it is removed by the output filter, not by the PWM.

Problem 5: SPWM Full-Bridge Inverter with Unipolar Voltage Switching

Problem 5: Unipolar Switching vs. Bipolar Output Filter Sizing

Statement: The inverter of Problem 4 is reconfigured for **unipolar voltage switching** with the same $V_{dc} = 300$ V, $m_a = 0.80$, $f_o = 50$ Hz and carrier frequency $f_s = 2$ kHz ($m_f = 40$, even, as required for unipolar switching).

1. Determine the peak and RMS fundamental output voltages.
2. What is the lowest significant harmonic frequency in the unipolar output voltage?
3. An LC low-pass filter with corner frequency $f_c = \frac{1}{2\pi\sqrt{LC}}$ is placed one decade below the lowest significant harmonic. Compare the required LC product for bipolar PWM (Problem 4) and unipolar PWM.

Solution:

1. **Fundamental Voltages** (identical to bipolar switching):

$$\hat{\mathbf{V}}_{\mathbf{o1}} = m_a V_{dc} = 0.80 \times 300 = \mathbf{240}\ \text{V}, \qquad \mathbf{V_{o1(rms)}} = \frac{240}{\sqrt{2}} = \mathbf{169.7}\ \text{V}$$

2. **Lowest Significant Harmonic:** all clusters at odd multiples of m_f cancel; the first cluster is around $2m_f = 80$, with components at $2m_f \pm 1$ ($0.314\,V_{dc} = 94$ V) and $2m_f \pm 3$ ($0.139\,V_{dc} = 42$ V); the $2m_f \pm 5$ lines ($0.013\,V_{dc}$) are negligible. The lowest significant one is

$$f_{\text{lowest}} = (2m_f - 3) f_o = 77 \times 50 = \mathbf{3850}\ \text{Hz}$$

3. LC **Filter Comparison:**

- **Bipolar** (Problem 4): the lowest significant harmonic is $(m_f - 2)f_o = 1850$ Hz. One decade below: $f_c = 185$ Hz:

$$(LC)_{\text{bipolar}} = \frac{1}{(2\pi \times 185)^2} = \mathbf{7.40 \times 10^{-7}}\ \text{s}^2$$

- **Unipolar:** lowest significant harmonic 3850 Hz; $f_c = 385$ Hz:

$$(LC)_{\text{unipolar}} = \frac{1}{(2\pi \times 385)^2} = \mathbf{1.71 \times 10^{-7}}\ \text{s}^2$$

Conclusion: the ratio is ≈ 4.3: doubling the frequency of the lowest harmonic cuts the LC product to about **one quarter** (each of L and C about halved), reducing the weight, copper loss and cost of the filter.

Problem 6: Three-Phase PWM Voltage Source Inverter

Problem 6: Three-Phase SPWM Line-to-Line Voltage Calculation

Statement: A three-phase bridge inverter is supplied from an electric vehicle battery pack of $V_{dc} = 400$ V. The inverter is controlled using three-phase sinusoidal PWM in the linear modulation range with $m_a = 0.85$ and $m_f = 21$.

1. Calculate the peak and RMS values of the fundamental phase-to-neutral (pole) voltage.
2. Calculate the peak and RMS values of the fundamental line-to-line output voltage.
3. Calculate the maximum achievable RMS fundamental line-to-line voltage in the linear range ($m_a = 1.0$), and compare it to the six-step square-wave mode.

Solution:

1. **Fundamental Phase Voltage (v_{Ao1}):**

$$\hat{\mathbf{V}}_{\mathbf{Ao1}} = m_a \frac{V_{dc}}{2} = 0.85 \times \frac{400}{2} = \mathbf{170.0}\ \text{V}$$

$$\mathbf{V}_{\mathbf{Ao1(rms)}} = \frac{\hat{V}_{Ao1}}{\sqrt{2}} = \frac{170}{\sqrt{2}} = \mathbf{120.2}\ \text{V}$$

2. **Fundamental Line-to-Line Voltage (V_{LL1}):** Using equations (56) and (57):

$$\hat{\mathbf{V}}_{\mathbf{LL1}} = \sqrt{3}\hat{V}_{Ao1} = \sqrt{3} \times 170.0\ \text{V} = \mathbf{294.4}\ \text{V}$$

$$\mathbf{V}_{\mathbf{LL1(rms)}} = \frac{\sqrt{6}}{4} m_a V_{dc} = 0.6124 \times 0.85 \times 400 = \mathbf{208.2}\ \text{V}$$

3. **Maximum Voltage Comparison:**

- **Maximum Linear SPWM ($m_a = 1.0$):**

$$V_{LL1(rms),\text{max}} = 0.6124 \times 400\ \text{V} = \mathbf{245.0}\ \text{V}$$

- **Six-Step Square-Wave Mode:**

$$V_{LL1(rms),\text{square}} = \frac{\sqrt{6}}{\pi} V_{dc} = \frac{\sqrt{6}}{\pi} \times 400 = 0.7797 \times 400 = \mathbf{311.9}\ \text{V}$$

Operating in six-step mode yields $\frac{311.9}{245.0} = 1.273$ (+27.3%) higher fundamental voltage than linear SPWM.

Problem 7: Third Harmonic Injection PWM (THIPWM) Voltage Boost

Problem 7: THIPWM DC Bus Utilization Advantage

Statement: A three-phase inverter fed from a $V_{dc} = 600$ V DC bus supplies a 415 V (RMS line-to-line), 50 Hz induction motor.

1. Can standard sinusoidal PWM in the linear range deliver 415 V?
2. What is the maximum line-to-line RMS voltage with third-harmonic-injection PWM while

staying in the linear range?

3. Can THIPWM supply the motor at rated voltage without over-modulation? What are the required m_a and the peak of the composite reference?

Solution:

1. **Standard SPWM:** the linear limit is $m_a = 1.0$:

$$V_{LL1(rms),\text{max (SPWM)}} = 0.6124 \times 600 = \mathbf{367.4}\ \text{V}$$

Since $367.4\ \text{V} < 415\ \text{V}$, SPWM cannot give rated voltage in the linear range (it would need $m_a = 415/367.4 = 1.13$, i.e. over-modulation with low-order harmonics).

2. **THIPWM:** the fundamental index may rise to $\frac{2}{\sqrt{3}} = 1.1547$:

$$V_{LL1(rms),\text{max (THIPWM)}} = 0.6124 \times 1.1547 \times 600 = \frac{V_{dc}}{\sqrt{2}} = \mathbf{424.3}\ \text{V}$$

3. **Feasibility:** $424.3\ \text{V} > 415\ \text{V}$, so THIPWM can supply the motor. The fundamental index required is

$$m_a = \frac{415}{0.6124 \times 600} = \mathbf{1.129}\ (\leq 1.1547)$$

Although $m_a > 1$, there is no over-modulation: the composite reference $\hat{V}_1\left[\sin\omega t + \frac{1}{6}\sin 3\omega t\right]$ peaks at $0.866 \times 1.129 = \mathbf{0.978}$ of the carrier amplitude, which is below 1. The motor voltage therefore stays sinusoidal and free of low-order harmonics, with a 2.2% voltage margin.

Problem 8: Analytical Evaluation of Dead-Time (Blanking Time) Voltage Error

Problem 8: Dead-Time Induced Fundamental Voltage Reduction

Statement: A three-phase IGBT inverter operates from a $V_{dc} = 750$ V DC bus at a switching frequency $f_s = 10$ kHz with a programmed blanking time $t_d = 2.5\,\mu$s.

1. Calculate the average pole-voltage error ΔV_{Ao} of one phase leg.
2. Determine the percentage reduction of the fundamental voltage at modulation index $m_a = 0.20$ (low-speed operation), assuming the phase current is in phase with the fundamental voltage.
3. Propose a compensation method for a DSP-controlled drive.

Solution:

1. **Average pole-voltage error:** using equation (49),

$$|\Delta V_{Ao}| = f_s\, t_d\, V_{dc} = (10 \times 10^3)(2.5 \times 10^{-6})(750) = 0.025 \times 750 = \mathbf{18.75}\ \text{V}$$

i.e. $t_d/T_s = 2.5\%$ of V_{dc}, with a sign opposite to the current: -18.75 V for $i_A > 0$ and $+18.75$ V for $i_A < 0$.

2. **Effect on the fundamental at $m_a = 0.20$:** the nominal peak fundamental pole voltage is

$$\hat{V}_{Ao1} = m_a \frac{V_{dc}}{2} = 0.20 \times \frac{750}{2} = 75.0 \text{ V}$$

The error is a square wave of amplitude 18.75 V in phase with $\text{sgn}(i_A)$; its fundamental has peak

$$\hat{V}_{\text{error},1} = \frac{4}{\pi}(f_s t_d V_{dc}) = \frac{4}{\pi} \times 18.75 = 23.87 \text{ V}$$

$$\text{Loss} = \frac{23.87}{75.0} \times 100\% = \mathbf{31.8}\%$$

Almost one third of the commanded voltage is lost at low speed. The error is a fixed voltage, so its relative effect grows as m_a falls; the square wave also contains 5th and 7th harmonics that produce torque pulsation. (With a lagging current the error vector is rotated with the current, adding a phase error as well.)

3. **Compensation:** the DSP senses the sign of each phase current and adds $+f_s t_d V_{dc} = 18.75$ V to the voltage reference of leg A when $i_A > 0$ (and -18.75 V when $i_A < 0$), equivalently shifts the PWM compare value by t_d. Near the current zero crossing the sign is uncertain, so practical schemes smooth the correction over a small current band.

Problem 9: Power Factor and THD in Non-Sinusoidal Rectifiers

Problem 9: True Power Factor and Distortion Factor Evaluation

Statement: A single-phase diode bridge rectifier with a capacitive filter draws a non-sinusoidal input current from a 230 V, 50 Hz supply. A power quality analyzer records the fundamental current component as $I_{s1} = 10$ A (RMS) lagging the line voltage by $\phi_1 = 18°$, and harmonic current components of: $I_{s3} = 7.5$ A, $I_{s5} = 4.5$ A, $I_{s7} = 2.0$ A, and $I_{s9} = 1.0$ A (higher harmonics negligible).

1. Calculate the total RMS input current I_s.
2. Compute the current Total Harmonic Distortion (THD_i).
3. Determine the Distortion Factor (K_{dist}) and Displacement Power Factor (DPF).
4. Calculate the True Power Factor (PF) and real active power drawn from the mains.

Solution:

1. **Total RMS Current (I_s):**

$$I_{\text{dist}}^2 = \sum_{h=3,5,7,9} I_{sh}^2 = (7.5)^2 + (4.5)^2 + (2.0)^2 + (1.0)^2 = 56.25 + 20.25 + 4.0 + 1.0 = 81.5 \text{ A}^2$$

$$I_s = \sqrt{I_{s1}^2 + I_{\text{dist}}^2} = \sqrt{10^2 + 81.5} = \sqrt{181.5} = \mathbf{13.47} \text{ A}$$

2. **Current Total Harmonic Distortion (THD_i):**

$$\text{THD}_\mathbf{i} = \frac{\sqrt{I_{\text{dist}}^2}}{I_{s1}} = \frac{\sqrt{81.5}}{10} = \frac{9.028}{10} = 0.9028 \implies \mathbf{90.28}\%$$

3. **Distortion Factor and Displacement Factor:**

$$\mathbf{K}_{\text{dist}} = \frac{I_{s1}}{I_s} = \frac{10\text{ A}}{13.47\text{ A}} = \mathbf{0.7424}$$
$$\text{DPF} = \cos\phi_1 = \cos(18^\circ) = \mathbf{0.9511}$$

4. **True Power Factor and Real Power:** Using equation (73):

$$\text{PF} = K_{\text{dist}} \times \text{DPF} = 0.7424 \times 0.9511 = \mathbf{0.706}\text{ (lagging)}$$

Verification using equation (76):

$$\text{PF} = \frac{\cos(18^\circ)}{\sqrt{1+(0.9028)^2}} = \frac{0.9511}{\sqrt{1+0.815}} = \frac{0.9511}{\sqrt{1.815}} = \frac{0.9511}{1.3472} = 0.706\text{ (Exact match!)}$$

Real active power consumed:

$$\mathbf{P} = V_s I_{s1}\cos\phi_1 = 230 \times 10 \times 0.9511 = \mathbf{2187.5}\text{ W} = \mathbf{2.188}\text{ kW}$$

Apparent power: $S = V_s I_s = 230 \times 13.47 = 3098$ VA. PF $= 2187.5/3098 = 0.706$.

Problem 10: Active Power Factor Correction (PFC) Boost Converter Design

Problem 10: Active PFC Boost Inductor Sizing & IEEE 519 Check

Statement: A 1.5 kW single-phase Active PFC boost converter is connected to a 230 V_{rms}, 50 Hz AC line. The regulated DC bus voltage is $V_d = 400$ V, switching frequency is $f_s = 100$ kHz, and target converter efficiency is $\eta = 95\%$.

1. Determine the peak and RMS fundamental input currents drawn from the AC line.
2. Calculate the maximum peak-to-peak inductor ripple current $\Delta I_{L,\text{max}}$ if ripple is limited to 20% of peak input current.
3. Calculate the required boost inductance L.
4. At the customer Point of Common Coupling (PCC), the available short-circuit current is $I_{sc} = 650$ A. What is the maximum permissible total demand distortion (TDD) specified by IEEE 519?

Solution:

1. **Input Line Currents:** Input active power drawn from the mains:

$$P_{\text{in}} = \frac{P_{\text{out}}}{\eta} = \frac{1500\text{ W}}{0.95} = 1578.9\text{ W}$$

Since the PFC converter shapes current to achieve near-unity power factor (PF ≈ 1.0):

$$\mathbf{I_{s(rms)}} = \frac{P_{\text{in}}}{V_s} = \frac{1578.9\text{ W}}{230\text{ V}} = \mathbf{6.86}\text{ A}$$
$$\hat{\mathbf{I}}_{\mathbf{s}} = \sqrt{2} \times I_{s(rms)} = \sqrt{2} \times 6.86 = \mathbf{9.71}\text{ A}$$

2. **Maximum Inductor Ripple Current ($\Delta I_{L,\max}$):**

$$\Delta I_{L,\max} = 20\% \times \hat{I}_s = 0.20 \times 9.71 \text{ A} = \mathbf{1.942} \text{ A}$$

3. **Boost Inductance Sizing (L):** In a boost PFC converter, the duty cycle varies continuously across the line cycle: $d(t) = 1 - \frac{|v_s(t)|}{V_d}$. The ripple current is:

$$\Delta i_L = \frac{|v_s| \cdot d}{f_s L} = \frac{|v_s|}{f_s L}\left(1 - \frac{|v_s|}{V_d}\right)$$

Differentiating with respect to $|v_s|$ shows that maximum ripple occurs precisely when $|v_s| = \frac{V_d}{2} = \frac{400}{2} = 200$ V (where duty cycle $d = 0.5$):

$$\Delta I_{L,\max} = \frac{V_d/2 \cdot 0.5}{f_s L} = \frac{V_d}{4 f_s L}$$

Rearranging to solve for inductance L:

$$\mathbf{L} = \frac{V_d}{4 f_s \Delta I_{L,\max}} = \frac{400 \text{ V}}{4 \times (100 \times 10^3 \text{ Hz}) \times 1.942 \text{ A}}$$
$$= \frac{400}{776,800} = 5.15 \times 10^{-4} \text{ H} = \mathbf{0.515} \text{ mH} \approx \mathbf{520}\,\mu\text{H}$$

4. **IEEE 519 TDD Limit Evaluation:** The maximum fundamental load current is $I_L = I_{s(rms)} = 6.86$ A. The Short-Circuit Ratio (SCR) at the PCC is:

$$\text{SCR} = \frac{I_{sc}}{I_L} = \frac{650 \text{ A}}{6.86 \text{ A}} = \mathbf{94.75}$$

From Table 7, for $50 < \frac{I_{sc}}{I_L} < 100$:

$$\text{Maximum Permissible TDD} = \mathbf{12.0}\%$$

Since active PFC achieves $\text{THD}_i < 5\%$, the installation effortlessly complies with IEEE 519 with massive safety margins!

Master Formula Sheet & Quick Revision Tables

Master Analytical Formula Matrix

Essential Examination Quick-Tips for KTU S5 EEE Students

1. **When asked for ACVC RMS Voltage Derivation:** Always state the half-wave symmetry condition ($v_o(\omega t + \pi) = -v_o(\omega t)$), integrate $[V_m \sin \omega t]^2$ from α to π (or α to β for RL load), convert $\sin^2\theta = \frac{1-\cos 2\theta}{2}$, and show every intermediate integration step clearly to secure full marks.

2. **Why $\alpha < \phi$ is prohibited in RL ACVC:** Remember the physical mechanism: If $\alpha < \phi$, the current in the previous thyristor is still flowing when the next thyristor is triggered. Conduction becomes continuous, phase control is lost ($V_{or} = V_s$ for every $\alpha < \phi$ with long gate pulses), and the thyristors behave as if the load were connected directly to the AC mains. With narrow gate pulses one thyristor may not fire at all, giving a unidirectional (DC-offset) current.

Table 9: Module 3 Master Analytical Formula Cheat-Sheet for Fast Exam Revision: ACVC and single-phase inverters

Topic / Circuit	Key Formulas / Governing Equations	Exam Tips & Critical Insights
ACVC (R Load)	$V_{or} = V_s\sqrt{1 - \frac{\alpha}{\pi} + \frac{\sin 2\alpha}{2\pi}}$ $\text{PF} = \frac{V_{or}}{V_s} = \sqrt{1 - \frac{\alpha}{\pi} + \frac{\sin 2\alpha}{2\pi}}$	At $\alpha = 0 \implies V_{or} = V_s, \text{PF} = 1.0$. At $\alpha = 90° \implies V_{or} = 0.707V_s, \text{PF} = 0.707$. At $\alpha = 180° \implies V_{or} = 0, \text{PF} = 0$.
ACVC (RL Load)	$i_o(\omega t) = \frac{V_m}{Z}[\sin(\omega t - \phi) - \sin(\alpha - \phi)e^{-\frac{R}{\omega L}(\omega t - \alpha)}]$ $\sin(\beta - \phi) = \sin(\alpha - \phi)e^{-\frac{R}{\omega L}(\beta - \alpha)}$ $V_{or} = V_s\sqrt{\frac{1}{\pi}[(\beta - \alpha) - \frac{\sin 2\beta - \sin 2\alpha}{2}]}$	**Rule:** $\alpha \geq \phi = \arctan(\omega L/R)$. If $\alpha = \phi \implies \beta = \pi + \alpha \implies \gamma = 180°$ (continuous sinusoidal current). Requires high-frequency pulse train gating!
ACVC (Integral Cycle)	$k = \frac{n}{n+m}, \quad V_{or} = V_s\sqrt{k}$ $\text{PF} = \sqrt{k}$	Zero switching EMI ($dv/dt = 0, di/dt = 0$). Causes line flicker/sub-harmonics. Used for thermal loads.
Half-Bridge VSI (Square Wave)	$v_o = \pm V_{dc}/2, \quad V_{or} = V_{dc}/2$ $\hat{V}_{o1} = \frac{2}{\pi}V_{dc}, \quad V_{o1(rms)} = \frac{\sqrt{2}}{\pi}V_{dc} \approx 0.45V_{dc}$	Requires 3-wire DC split supply. Switch voltage rating $= V_{dc}$ ($2\times$ peak output).
Full-Bridge VSI (Square Wave)	$v_o = \pm V_{dc}, \quad V_{or} = V_{dc}$ $\hat{V}_{o1} = \frac{4}{\pi}V_{dc}, \quad V_{o1(rms)} = \frac{2\sqrt{2}}{\pi}V_{dc} \approx 0.90V_{dc}$ $\text{THD}_v = \sqrt{\frac{\pi^2}{8} - 1} = 48.34\%$	Odd harmonics only ($n = 1, 3, 5, 7, \dots$). Harmonic amplitude $\hat{V}_{on} = \hat{V}_{o1}/n$. Twice the voltage, $4\times$ power of half-bridge!
Full-Bridge VSI, RL load	$I_0 = -\frac{V_{dc}}{R}\tanh\frac{T}{4\tau}, \quad t_0 = \tau\ln\left(1 + \tanh\frac{T}{4\tau}\right)$ $i_o = \sum\frac{4V_{dc}}{n\pi Z_n}\sin(n\omega t - \phi_n)$	Current is negative for $0 < t < t_0$ although $v_o = +V_{dc}$: carried by feedback diodes D_1, D_2. Sequence: $D_1D_2 \to S_1S_2 \to D_3D_4 \to S_3S_4$.
Single-Pulse PWM	$V_{or} = V_{dc}\sqrt{\delta/\pi}$ $\hat{V}_{o1} = \frac{4V_{dc}}{\pi}\sin\frac{\delta}{2}, \quad v_{on} \propto \sin\frac{n\delta}{2}$	$\delta = 120°$ removes the 3rd, 72° the 5th harmonic. Only one harmonic can be eliminated at a time.
Single-Phase Bipolar SPWM	$m_a = \frac{\hat{V}_{\text{control}}}{\hat{V}_{\text{tri}}}, \quad m_f = \frac{f_s}{f_o}$ $\hat{V}_{o1} = m_aV_{dc}, \quad V_{o1(rms)} = \frac{m_aV_{dc}}{\sqrt{2}}$	m_f must be an odd integer (eliminates even harmonics). Lowest harmonics clustered at $m_f \pm 2$.
Single-Phase Unipolar SPWM	$\hat{V}_{o1} = m_aV_{dc}, \quad V_{o1(rms)} = \frac{m_aV_{dc}}{\sqrt{2}}$ Effective switching frequency $= 2f_s$	3-level output ($+V_{dc}, 0, -V_{dc}$). Harmonics around m_f cancel! First cluster at $2m_f$. LC product reduced to about $\frac{1}{4}$ (L, C each $\approx$ halved).

Table 10: Module 3 Master Analytical Formula Cheat-Sheet (continued): three-phase inverters, dead time, harmonics and PFC

Topic / Circuit	Key Formulas / Governing Equations	Exam Tips & Critical Insights
Three-Phase SPWM (Linear)	$\hat{V}_{Ao1} = m_a \frac{V_{dc}}{2}$ $\hat{V}_{LL1} = \frac{\sqrt{3}}{2} m_a V_{dc} \approx 0.866 m_a V_{dc}$ $V_{LL1(rms)} = \frac{\sqrt{6}}{4} m_a V_{dc} \approx 0.6124 m_a V_{dc}$	Triplen harmonics (3rd, 9th, 15th, ...) cancel in line-line! Choose m_f as an odd multiple of 3 $(9, 15, 21, \dots)$. Max linear RMS line voltage $= 0.6124 V_{dc}$.
Three-Phase Six-Step (Square)	$V_{LL(rms)} = \sqrt{\frac{2}{3}} V_{dc} \approx 0.8165 V_{dc}$ $V_{LL1(rms)} = \frac{\sqrt{6}}{\pi} V_{dc} \approx 0.7797 V_{dc}$	180° conduction. Output contains $n = 6k \pm 1$ $(5th, 7th, 11th)$. Fundamental is 27.3% higher than max linear SPWM.
Third Harmonic Injection (THIPWM)	$v_{\text{control}} = \hat{V}_1[\sin(\omega t) + \frac{1}{6}\sin(3\omega t)]$ $V_{LL1(rms),\max} = \frac{V_{dc}}{\sqrt{2}} \approx 0.707 V_{dc}$	Composite reference peaks at only $0.866\hat{V}_1$ (at $60^\circ, 120^\circ$), so m_a may reach 1.1547. **Boosts linear output voltage by 15.5%!** Third harmonic cancels out in line-to-line output!
Blanking Time (Dead-Time)	$\lvert\Delta V_{Ao}\rvert = \frac{t_d}{T_s} V_{dc} = f_s t_d V_{dc}$ per leg, sign opposite to i_A Single-phase bipolar bridge: $\lvert\Delta V_o\rvert = 2 f_s t_d V_{dc}$ Fundamental of error: $\frac{4}{\pi} f_s t_d V_{dc}$ (peak, per leg)	Prevents DC-link shoot-through. Pulse shorter by t_d for $i_o > 0$, longer for $i_o < 0$. Worst at low m_a; causes 5th/7th harmonics.
Harmonics & Power Factor	$\text{PF} = \left(\frac{I_{s1}}{I_s}\right) \cos\phi_1 = \frac{\cos\phi_1}{\sqrt{1+\text{THD}_i^2}}$ $\text{THD}_i = \frac{\sqrt{I_s^2 - I_{s1}^2}}{I_{s1}}$	True PF = Distortion Factor $\times$ Displacement Factor. Harmonics alone reduce power factor even if $\cos\phi_1 = 1.0$!
Active PFC Boost Inductor	$\Delta I_{L,\max} = \frac{V_d}{4 f_s L} \implies L = \frac{V_d}{4 f_s \Delta I_{L,\max}}$	Max ripple occurs at $d = 0.5$ $(\lvert v_s\rvert = V_d/2)$. Universal AC input, regulates $V_d \approx 400$ V DC, $\text{THD}_i < 5\%$.

3. **Fourier Series of Square Wave Inverter:** Write down the general half-wave symmetry property first. State why all even harmonics and DC term are zero ($a_0 = 0, a_n = 0, b_{\text{even}} = 0$). Write the general formula $\sum \frac{4V_{dc}}{n\pi} \sin(n\omega t)$ and calculate $\text{THD}_v = \sqrt{\frac{\pi^2}{8} - 1} = 48.34\%$.

4. **Bipolar vs. Unipolar PWM Comparison:** Draw the two-level vs. three-level output waveforms side-by-side. Highlight the three killer facts: (1) Effective switching frequency doubles to $2f_s$; (2) Harmonics around m_f cancel out completely; (3) The LC product of the output filter falls to about $\frac{1}{4}$ (L and C each about halved).

5. **Why Triplen Harmonics Disappear in 3-Phase Inverters:** State that triplen harmonics (3rd, 9th, 15th, ...) form a zero-sequence set—they are exactly equal in magnitude and in phase in all three inverter legs ($v_{Ao,3} = v_{Bo,3} = v_{Co,3}$). When computing line-to-line voltage $v_{AB} = v_{Ao} - v_{Bo}$, they subtract to exactly zero!

6. **Dead-Time Derivation:** Draw the leg diagram showing S_1, S_4 and diodes D_1, D_4. Explain that when both switches are off during t_d, the load current i_o forces either D_1 or D_4 to conduct depending solely on the sign of i_o, subtracting or adding $V_{dc}t_d$ of volt-seconds to the output pulse. The average pole-voltage error is $\frac{t_d}{T_s}V_{dc} = f_s t_d V_{dc}$ (not $4f_s t_d V_{dc}$; the $\frac{4}{\pi}$ appears only in the fundamental of the error square wave).

7. **IEEE 519 PCC Rule:** Remember that IEEE 519 applies strictly at the **Point of Common Coupling (PCC)**, NOT inside the customer plant. Limits are specified in terms of **Total Demand Distortion (TDD)** and depend directly on the short-circuit ratio I_{sc}/I_L.

Short-Answer Drill (Part-A style, 3 marks each)

1. **State the RMS output voltage of a single-phase full-wave ACVC with R load.** $V_{or} = V_s\sqrt{1 - \frac{\alpha}{\pi} + \frac{\sin 2\alpha}{2\pi}}$; it falls from V_s at $\alpha = 0$ to $0.707V_s$ at 90° and 0 at 180°.

2. **Why is a high-frequency pulse train used to gate the thyristors of an RL ACVC?** If the first pulse arrives while the load current has not yet fallen to zero (or while the thyristor is not yet forward-biased), the thyristor would not latch; a pulse train keeps re-triggering until it does.

3. **Define m_a and m_f.** $m_a = \hat{V}_{control}/\hat{V}_{tri}$ sets the fundamental, $\hat{V}_{o1} = m_a V_{dc}$ for $m_a \leq 1$; $m_f = f_s/f_o$ sets where the harmonics appear.

4. **Why must m_f be odd (bipolar) and odd and a multiple of 3 (three-phase)?** Odd m_f gives half-wave symmetry and removes even harmonics; a multiple of 3 makes the m_f harmonic identical in the three legs so it cancels in the line voltage.

5. **What are the advantages of unipolar over bipolar switching?** Three output levels, the output pulse frequency is $2f_s$, harmonics around m_f are cancelled (first cluster at $2m_f$), current ripple and filter size are smaller, and the dv/dt step is V_{dc} instead of $2V_{dc}$.

6. **Why is blanking time needed, and what is its drawback?** It prevents shoot-through of the two switches of a leg ($t_{off} > t_{on}$). It makes the pole voltage depend on the current sign, so the output is reduced by about $f_s t_d V_{dc}$ per leg and low-order harmonics appear.

7. **What happens in the over-modulation region?** $m_a > 1$: pulses are dropped, $\hat{V}_{o1}$ grows non-linearly towards $\frac{4V_{dc}}{\pi}$ and low-order harmonics appear.

8. **How does third-harmonic injection raise the output voltage?** Adding $\frac{1}{6}\sin 3\omega t$ lowers the peak of the reference to 0.866 of its fundamental, so the fundamental can be raised by 1.1547 (15.5%) with no over-modulation. The third harmonic is zero-sequence and cancels in the line voltage.

9. **Compare a VSI and a CSI.** A VSI has a stiff DC voltage source: output voltage imposed by the switches, current by the load, anti-parallel diodes needed, shoot-through is fatal, an open circuit is harmless. A CSI has a stiff DC current source (L_{dc}): output current imposed by the switches, voltage by the load, series blocking diodes needed, simultaneous conduction in a leg is harmless, an open circuit is fatal.

10. **Define displacement factor, distortion factor and power factor of a non-sinusoidal current.** DPF $= \cos\phi_1$; distortion factor $= I_{s1}/I_s = 1/\sqrt{1+\text{THD}_i^2}$; PF $= I_{s1}\cos\phi_1/I_s$.

11. **What does IEEE 519 limit, and where?** Harmonic current (as TDD, with limits depending on I_{sc}/I_L) and harmonic voltage (THD$_v$ and individual) at the PCC.

12. **How does an active PFC boost converter shape the input current?** An outer slow loop regulates V_d; its output multiplies $|v_s|$ to give $i_L^* = Kv_e|v_s|$; a fast inner loop forces the boost inductor current i_L to follow i_L^*, so the input current is a sinusoid in phase with the line voltage.